MATH INQUIRERS HOLIDAY PROGRAM

Math Inquirers P6 Tutor Guide

Holiday program for the Primary 6 level

Notice it. Test it. Explain why.

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Program guide for tutors: schedule, lesson routine, ten minute-by-minute lesson scripts with answer keys, a progress tracker and a parent summary. Math Inquirers is online first: every lesson runs live on video; the face-to-face notes are kept for partner centres.

The program at a glance

Level	Primary 6 students after the PSLE (ages 12), after the PSLE, going into Secondary 1, who are ready for more thinking than standard practice
Length	10 lessons of 90 minutes, two a week over the 5 weeks of the holidays
Group size	Up to 8 children per tutor, live online. Pilot with 6. Add a second tutor or helper above 8.
Focus	Thinking skills: making a list in order, guess and check, drawing a model, working backwards, finding patterns, reasoning and explaining. Not speed drills.
The arc of every lesson	Start concrete with objects, record in a table, spot the pattern, state the rule, use it (often backwards), then explain why.
What children take home	A student workbook with an inquiry journal, 10 worksheets and a certificate
Before tutors start	Every tutor signs the tutor confidentiality agreement before receiving any materials.

Schedule

Week	Lesson	Title	Topic	Big idea
Week 1a	1	**Code the Pattern**	Sequences and the nth Term	Pattern spotting
Week 1b	2	**Prime Hunters**	Primes, Factors and Prime Factorisation	Systematic listing (factor pairs)

Week 2a	3	**Below Zero**	Negative Numbers	Number line reasoning
Week 2b	4	**Algebra Machines**	Expressions and Equations	Rule finding from a table
Week 3a	5	**Every Possible Way**	Counting	Systematic listing
Week 3b	6	**Fair or Unfair?**	Probability	Listing equally likely outcomes
Week 4a	7	**Turtle Geometry**	Angles in Polygons	Turn (exterior) and inside (interior) angles
Week 4b	8	**Secret Codes**	Clock (Modular) Arithmetic	Remainders and mod
Week 5a	9	**Data Detectives**	Averages and Data	Mean
Week 5b	10	**Showcase Challenge**	Mixed: Maths and Code	Choosing a strategy

Suggested pairing: two lessons a week, for example Tuesday and Thursday, or Saturday and Sunday. Keep the same time slot all five weeks.

The 90-minute lesson routine

Every lesson uses the same routine so children know what to expect and tutors can prepare quickly.

Minutes	Part	What happens	Face to face	Online
0–10	**Warm-up**	A quick game from the lesson plan. Everyone joins in.	Sit in a circle or horseshoe.	Cameras on. Answers by thumbs, hands or chat.
10–25	**Explore**	Hands-on in pairs. Objects first, no writing yet.	Materials box on each table.	Each child has their own materials at home. Pairs go to breakout rooms.
25–35	**Mini-lesson**	Tutor builds the table on the board, names the rule and asks what stays the same and what changes.	Whiteboard.	Shared slide and annotation tool.
35–40	**Brain break**	Stand, stretch, drink water.	A short movement game.	Stand up on camera. Count the stretches together.
40–75	**Worksheet**	Start in pairs, finish alone. Tutor asks “how do you know?” rather than giving answers.	Printed workbook, tutor circulates.	Printed workbook. Child holds the page up to the camera. Tutor checks one or two answers each.
75–82	**Talk time**	Share one answer and explain it. Compare methods.	Each child shares in turn.	Each child unmutes in turn.
82–90	**Exit ticket and stamp**	One question on a slip of paper. Stamp the inquiry journal.	Paper slip, tutor stamps.	Child writes the answer and shows the camera. Tutor stamps the next time or posts it.

Teaching habits that matter

- Ask “How do you know?” after every answer, right or wrong.
- Wait 10 seconds before helping. Let children struggle a little.
- Praise the thinking (“You checked it two ways”), not just the answer.
- Let children disagree with each other and settle it with objects or a drawing.
- If a child finishes early, use the Challenge or Extension question, not extra drill.

Preparing the materials

- Send the workbook PDF to parents one week before. Ask them to print it single-sided and the cut-out pages on card if they can.
- Each child needs a computer or tablet with Scratch 3 (free online). Python is optional (any online Python editor). Cut-outs used: unplugged code cards (Lessons 1, 4, 7), square grid (Lessons 7, 9).

- Keep the answer key with the tutor only. Check answers together on screen.
- Use the shared screen or a digital whiteboard for the mini-lesson table.

Setting up online

- Send parents the workbook and materials list one week before. Needed: the printed workbook and cut-outs, scissors, a ruler, paper and a pencil, and a computer with Scratch.
- Use one approved platform with breakout rooms and screen share. Admit only registered children and never share the link publicly.
- Ask the parent to sit nearby but not give answers. The camera should show the child's page and hands.
- Hold up the answer page to the camera at the end of each lesson, or take a photo and send it through the approved channel.
- Do not record lessons (this is in the tutor agreement). Add a second tutor or helper above 8 children.

Materials master list

Lesson	Title	Face to face	Online set-up
1	Code the Pattern	Laptop or desktop with Scratch 3 open (scratch.mit.edu, no account needed), about 25 matchsticks or toothpicks per child (or squared paper), mini whiteboard, workbook. Optional: a Python editor for children who already know Python.	The tutor shares the slides and a Scratch window. Share your Scratch window and build the four blocks together on screen; the blocks are also on the slide. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. Scratch is free at scratch.mit.edu and needs no account in class. Children hold paper up to the camera or share their screen when asked. Matchsticks can be drawn as lines.
2	Prime Hunters	Laptop or desktop with Scratch 3 open, a printed 1 to 100 grid (or squared paper), two coloured pencils, mini whiteboard, workbook. Optional: a Python editor for children who already know Python.	Share the 1 to 100 grid on screen and annotate it while children work on their own printed or hand-drawn grid. Build the prime checker together in your shared Scratch window; the blocks are also on the slide. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. Children hold their grid up to the camera at minute 22.

3	Below Zero	Laptop or desktop with Scratch 3 (scratch.mit.edu, free, no log-in needed), mini whiteboard and marker, workbook, pencil.	Children keep Scratch open in one browser tab and the video call in another. The tutor screen-shares Scratch to show the x number under the stage. Children share their screens only when asked; page 1 of the worksheet is done on paper with laptops turned away. Type the minus sign with the keyboard minus key (the hyphen); do not copy and paste a number from the slide or chat, because the slide uses a different minus character that Scratch will not accept. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour.
4	Algebra Machines	Laptop or desktop with Scratch 3 (scratch.mit.edu, free), mini whiteboard and marker, workbook, pencil.	Pairs work in breakout rooms for the explore: one partner shares their screen with the machine running and types in the numbers the other partner calls out. Page 1 of the worksheet is done on paper with laptops turned away. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour.
5	Every Possible Way	Laptop or desktop with Scratch 3 (scratch.mit.edu) open, the starter project “Every Order” (one loop that says each tea), mini whiteboards, digit cards 0 to 9, squared paper.	Children share their screen when they test code. The tutor shares the starter project as an .sb3 file in the chat (children use File, Load from your computer; no account needed), and the three starter blocks are also on the slide. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. Digit cards can be cut from paper. Breakout rooms in pairs for the Explore.

6	Fair or Unfair?	Laptop or desktop with Scratch 3 (scratch.mit.edu), two dice per pair (or the tutor's dice program), a coin, coloured counters and a paper bag, mini whiteboards, tally sheet.	Children roll real dice at home; anyone without dice builds the dice program live in the shared Scratch window (pick random 1 to 6, twice) and uses that. This lesson is enrichment: probability is new to most children, so keep it playful. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. Pairs play the Explore game in breakout rooms and type their tallies into the chat so the group total can be added up.
7	Turtle Geometry	A laptop or desktop with Scratch 3 open (scratch.mit.edu) and the Pen extension added, mini whiteboards, a protractor, plain or squared paper.	Pairs work in breakout rooms: one child drives Scratch and shares the screen, the other predicts, then they swap. Children hold drawings up to the camera. A tablet works for Scratch but is slower. Every drawing starts with the same 4 set-up blocks (erase all, go to x: -50 y: 100, point in direction 90, pen down) so shapes stay on the stage; turn blocks in Scratch show an arrow (turn ↻ is clockwise, our "turn right"). Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left (Motion, Pen, Control), not by colour.

8	Secret Codes	A laptop or desktop with Scratch 3 open (the tutor shares a starter project with a list called alphabet holding A to Z), mini whiteboards, paper for code strips, scissors.	Children make their code strips on paper and hold them to the camera. Children hold up their coded name on paper to the camera, or the tutor reads out the coded names; partners decode on paper. Pairs use breakout rooms for the coding part; the driver shares the screen. Starter project: the tutor shares an .sb3 file (not a link only) containing the list alphabet with A to Z already filled in, and the loop already built; children add the three cipher blocks. Block colours on our slides are simplified; in Scratch, list blocks are red-orange (Variables palette) and you find blocks by the category name on the left, not by colour.
9	Data Detectives	Laptop or desktop with Scratch 3 (scratch.mit.edu; free, in the browser), workbook, pencil, mini whiteboard, about 25 linking cubes or counters for levelling towers. A calculator only for checking.	Data is collected in the chat. Children share their screen to show their code. The tutor shares a starter project as an .sb3 file (one sprite with an empty list called data; children use File, Load from your computer) or children build it from the slide in 5 minutes. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. The mean is familiar from primary school; the new parts are lists and loops, median and mode, and misleading averages.

10	Showcase Challenge	Laptop or desktop with Scratch 3 (scratch.mit.edu), workbook (challenge round on page 1, planning sheet on page 2), pencil, mini whiteboard, the starter projects as .sb3 files (A Mystery Machine, B Prime or Not?, C Lucky Seven, D Clock Jumper; children use File, Load from your computer), or build them from the slides, certificates, stamps.	Challenge round in teams of 2 in breakout rooms. Building: everyone in the main room with cameras on, asking for help in the chat. Showcase: each child shares their screen for 3 minutes; the group gives two stars and a wish in the chat. Children may paste a screenshot into the chat instead and the tutor runs their file, which removes screen-share switching. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour.

How to use the scripts

Every lesson below is a minute-by-minute script. The slides match it: one slide or two for each part. You do not have to say the words exactly, but keep the questions, because the questions are what makes children think.

SAY	Words to say aloud, in plain language for 8- and 9-year-olds. Said in your own voice, not read from the page.
DO	What you do: set up, hand out, show a slide, or move around the room.
ASK / EXPECT	Questions to put to the group, with the answer you are listening for. Let children answer before you explain.
IF STUCK	A ladder of help. Give hint 1, wait ten seconds, then give hint 2. Do not give the answer.

Five habits that run through every script

- Predict first. Before children test anything, ask for a guess and write it down. A wrong guess is useful because we can test it.
- Wait ten seconds after every question. Count silently. Most children answer in the last four.
- Ask “How do you know?” after right answers too, so that children do not read your face for the answer.
- Praise the thinking, not the answer: “You checked it two ways” beats “Well done”.
- When an answer is wrong, say “Let us test that,” and use objects, not “No”.

Answering a wrong answer

If...	Then...
A child gives a wrong answer	“Thank you. Let us test it with the counters.” Then build it together.
A child gives a right answer but cannot explain	“Show me with your counters.” Then: “Tell your partner how you know.”
A child says “I do not know”	“What do you notice? What could you try first?” Offer hint 1.
Two children disagree	“Both of you, convince the group using objects.” Let the objects decide.

Lesson 1: Code the Pattern

Week 1 (first lesson) • Sequences and the nth Term • Workbook: Lesson 1

Objective	Continue a sequence, find the rule for the nth term of a sequence with a constant difference, use it to find far terms such as the 100th, work backwards to decide whether a number is in a sequence, and test predictions with a loop and a variable in Scratch.
Thinking skills	Pattern spotting, generalising with a letter, predicting, testing in code, debugging (counting the jumps)
Materials	Laptop or desktop with Scratch 3 open (scratch.mit.edu, no account needed), about 25 matchsticks or toothpicks per child (or squared paper), mini whiteboard, workbook. Optional: a Python editor for children who already know Python.
Online set-up	The tutor shares the slides and a Scratch window. Share your Scratch window and build the four blocks together on screen; the blocks are also on the slide. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. Scratch is free at scratch.mit.edu and needs no account in class. Children hold paper up to the camera or share their screen when asked. Matchsticks can be drawn as lines.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Welcome to Math Inquirers P6! The PSLE is done, so there is no exam today: just puzzles, patterns and code."*

"Every lesson we notice a pattern, test it in code, and explain why. Today's question: how can you find the 100th number in a pattern without writing out all 100?"

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

2–10 min Warm-up

DO Warm-up check first (slide 2, 3 minutes). It comes out of the 10-minute warm-up: play the game for about 3 minutes and ask only the first three quick questions after it.

SAY *"Two quick puzzles. They are new, not from last time. Write your answer AND one reason on your whiteboard. Hold it up when I say."*

Warm-up check	Answer and the reason to listen for
A taxi charges \$3 to start plus \$2 for every km. The fares are \$5 for 1 km, \$7 for 2 km and \$9 for 3 km. What is the fare for 20 km?	\$43. Each km adds \$2, so the fare is 2 × 20 + 3 = 43. (Not 2 × 20 = 40: the \$3 start charge is there from the beginning.)
Is 50 a term of 7, 12, 17, 22, …? How do you know?	No. The rule is 5n + 2. 50 − 2 = 48 is not a multiple of 5. (Every term ends in 2 or 7, and 50 ends in 0.)

DO Fast track (nobody is labelled). If a child has BOTH answers right and a sensible reason, send a private chat message: “Today start on the Think Harder page at minute 40.” Fast-track children still join Explore, the mini-lesson and Talk time; only the worksheet time changes. Tell everyone: “Some of you start on a different page today. It changes every lesson, and you can move any time.” Never say “advanced”, never record the check as a mark, and let any child choose the main pages. A fast-track child who gets stuck goes back to the main page. This works because groups are small (up to 8) and each child has a printed workbook, so you can give children different pages quietly.

DO Next! Pattern quick-fire. Six sequence cards. Teams race to say the next two terms; a bonus point for the rule.

SAY *“Teams of two. I will show a sequence card. The first team to say the next two terms gets a point, and a bonus point if you can say the rule.”*

“Careful: not every card adds the same amount each time.”

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
4, 9, 14, 19, __	24
The 10th term of 2, 4, 6, 8, …	20
The 10th term of 3, 5, 7, 9, …	21
Ali has $12 and saves $5 a week. How much does he have after 6 weeks?	$42
100, 93, 86, __	79

10–25 min Explore (in pairs, with objects)

SAY *“Pattern 1 is a triangle made of 3 matchsticks. Each new pattern adds one more triangle in the row. Before you build anything: how many sticks will pattern 100 need? Write your guess in the corner of your whiteboard.”*

DO 1. Build matchstick triangles in a row: pattern 1 has 1 triangle, pattern 2 has 2, pattern 3 has 3.

2. Count the sticks in each pattern. Make a table.

3. Predict the sticks in pattern 10 and in pattern 100. Do not build them!

4. Check pattern 10 in Scratch with a loop and a variable.

SAY *“Talk to your partner while you work. I will come round and ask you questions.”*

IF STUCK Stuck building: lay the triangles point up, point down, point up, so each new triangle shares a stick.

Stuck on pattern 100: “How many sticks does each new triangle add? How many new triangles come after the first one?”

Finished early: in Scratch, set sticks to 3, repeat 9, change sticks by 2, say sticks. Does it match your pattern 10? Now change it to reach pattern 100.

DO At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How many sticks are in pattern 10?	21
How many in pattern 100? How did you get it without building it?	201. Each new triangle adds 2 sticks: 3 + 2 × 99 = 201, or 2 × 100 + 1.
Which pattern uses exactly 75 sticks?	Pattern 37: 2 × 37 + 1 = 75.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"5, 8, 11, 14. The term-to-term rule is add 3. That is great for the next term, but slow for the 100th: I would have to add 3 ninety-nine times."*

"Adding 3 each time is like the 3 times table: 3, 6, 9, 12. Each term is 2 more than the times-table number. So the nth term is 3 × n + 2, which we write 3n + 2. Check the 4th term: 3 × 4 + 2 = 14. Yes."

"So the 100th term is 3 × 100 + 2 = 302. Now I test it in code. I set term to 5, the 1st term. From the 1st term to the 100th term there are 99 jumps, so repeat 99, change term by 3, then say term. It says 302. The maths and the code agree."

Worked example on the slides	Answer
The 100th term of 5, 8, 11, 14, ... The difference is 3, so compare with the 3 times table. Each term is 2 more: nth term = 3n + 2. Check the 4th term: 3 × 4 + 2 = 14.	3 × 100 + 2 = 302.
Test it in code. set term to 5 (the 1st term). repeat 99: change term by 3 (99 jumps from the 1st term to the 100th). say term.	It says 302, the same as the rule.

SAY *"The key idea today is: the rule for the nth term. The difference tells you the times table. Compare with it to find what to add or take away. Say it with me."*

DO Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Using only the difference as the rule (writing 3n for 5, 8, 11, ...) and not comparing with the 1st term.
- Off by one: doing 100 jumps from the 1st term (5 + 3 × 100) instead of 99, in the maths and in the repeat block.

- Putting "change" before "say" in a loop, so the sprite never says the 1st term.

35–40 min Brain break

DO Pattern moves: clap, clap, stomp, clap, clap, stomp … Do 12 moves together. Then predict: what is the 20th move? The 30th?

Water. Toilet break for anyone who needs it. Keep it to five minutes.

40–75 min Worksheet (pairs, then alone, then stretch)

DO Minute 40: give out the workbook. Read the title of Lesson 1 aloud. Say “Pairs first. Talk before you write.”

Minutes 40 to 55: pairs work together on the first page (with the short teaching slides). Minutes 55 to 72: each child works alone on the second page. Minutes 72 to 75: check together, then stretch and challenge for early finishers.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: “What is the difference? Which times table is it like?”
- Ask: “Check your rule with the 1st term AND the 4th term. Does it work for both?”
- Ask: “How many jumps are there from the 1st term to the 20th term?”

IF STUCK Hint 1: “Write the times table of the difference under the sequence.”

Hint 2: “How far is each term from the times-table number under it?”

Hint 3: “Count the jumps for a small case first: from the 1st term to the 4th term is 3 jumps.”

DO Early finishers: give the stretch question first, then the challenge, then the Open investigation (it is the last box on the second Think Harder page). Fast-track children start there at minute 40.

Stretch and challenge questions

Question	Answer
Workbook Think Harder 1: Find the 100th term of 12, 19, 26, 33, …	705. The rule is 7n + 5, and 7 × 100 + 5 = 705.
Workbook Think Harder 2: Is 200 a term of 2, 9, 16, 23, …? Explain.	No. The rule is 7n − 5. The 29th term is 198 and the 30th term is 205, so 200 is skipped.
Workbook Think Harder 3: Matchstick houses in a row: 1 house uses 6 sticks, 2 houses use 11, 3 houses use 16. How many houses can you make with 100 sticks?	19 houses. The rule is 5n + 1: 19 houses use 96 sticks (4 left over) and 20 houses would need 101.
Workbook Think Harder 4: set n to 7, then repeat 49 [change n by 6], then say n. What does it say? Which term of 7, 13, 19, … is that? Check with the rule 6n + 1.	301, the 50th term (7 + 49 × 6 = 301, and 6 × 50 + 1 = 301). Repeat 49 reaches the 50th term, not the 49th.

Workbook Challenge 5: Make up your own sequence puzzle. Choose a first term and a difference, write the first four terms, then ask: "Is ___ a term?" Pick a number that is NOT a term but is close to one. Swap with a partner and explain how you know.	Open. Example: 6, 11, 16, 21, … (rule 5n + 1). Is 100 a term? No: 100 − 1 = 99 is not a multiple of 5 (the 19th term is 96 and the 20th is 101). Look for a reason from the rule or from counting jumps, not just a list.
Workbook Challenge 6: Which numbers below 60 are in BOTH sequences 2, 5, 8, 11, … and 4, 9, 14, 19, …? Find them all.	14, 29, 44 and 59. One more than each of them is a multiple of both 3 and 5, so a multiple of 15 (15, 30, 45, 60).
Workbook Challenge 7: What does this code say? set t to 0, set k to 0, then repeat 5 [change k by 1, change t by k], then say t. What would it say with repeat 100?	15 (1 + 2 + 3 + 4 + 5). With repeat 100 it says 5050: pair 1 + 100, 2 + 99, … to make 50 pairs of 101.
Workbook Open investigation (What if we change …? What stays the same?): Triangles in a row use 3, 5, 7, … sticks (the rule is 2n + 1). What if we join squares in a row, or pentagons, or hexagons, sharing one side each time? Pick a shape. Draw patterns 1, 2 and 3, make a table and find the rule. What changed? What stayed the same?	Regular shapes with k sides joined along one side each time use (k − 1) × n + 1 sticks: triangles 2n + 1, squares 3n + 1, pentagons 4n + 1, hexagons 5n + 1. What changes: the number added each time, which is always one less than the number of sides because a side is shared. What stays the same: the rule has the form (something) × n + 1, and pattern 1 is the single shape (k sticks). Look for: a table, the difference named, a check of the rule on pattern 1 and one more pattern, and a reason for the sharing. Any shape chosen is valid. Stretch: a child may write the rule for any k, or test it in code (set s to k, repeat n − 1 [change s by k − 1]).

These are the questions on the two Think Harder pages in the workbook. More are in the separate Question Bank.

75–82 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask two or three children to share, not only the fastest.

SAY *"Ali says the 100th term of 7, 10, 13, … is 7 + 3 × 100 = 307. Is he right?"*

"When is the nth-term rule better than adding on? When is code better than both?"

Discussion question	Listen for
Ali says the 100th term of 7, 10, 13, … is 7 + 3 × 100 = 307. Is he right? Explain with a rule, a picture or code.	No. The 1st term is already 7 before any jump, so the 100th term is only 99 jumps away: 7 + 3 × 99 = 304. The rule 3n + 4 gives 304 too. The rule is fastest for one far term; code is best for listing many terms or checking an answer.

82–90 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
The sequence 8, 11, 14, 17, … (a) Write the rule for the nth term. (b) Find the 100th term. (c) A loop starts at the 1st term. How many times must it repeat "change by 3" to reach the 100th term?	(a) 3n + 5 (b) 305 (c) 99 times

SAY *"Today you found the 100th term without writing 100 numbers, and you tested it in code. That is what mathematicians and programmers both do."*

"Do the exit ticket on your own, then show me."

Reflection: What was hard? (1 minute)

SAY *"Last minute, just for you. On the last page of today's lesson, finish the three sentences in the box called "What was hard?": the hardest part was, I got unstuck by, next time I will try. Nobody marks it, and nobody has to show it. Then I would like to hear ONE sentence from two or three of you. "Pass" is fine."*

DO Run it at about minute 88, while you collect exit tickets (slide at the end). Never grade or correct it, and never read it out without the child's say-so. Praise the strategy, not the speed: say "you you counted the jumps — that is what got you unstuck", never "you were quick". If a child writes "nothing was hard", ask "what would make it harder?".

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Extension for children who are ready

Make the program say the position and the term together (for example "term 7 is 23") using a second variable pos and the join block. Then make it stop at the first term bigger than 1000.

Answer key (workbook)

Q1 (a) 4, 7, 10, 13, 16, 19. Each new square adds 3 sticks, not 4, because it shares one side with the square before it. (b) sticks = 3 × n + 1 (c) 301 (d) Pattern 20 (3 × 20 + 1 = 61)

Q2 (a) difference 4; 4n + 2; 402 (b) 5; 5n − 4; 496 (c) 2; 2n + 7; 207 (d) 0.5; 0.5n (or n ÷ 2); 50

Q3 (a) 4n + 1 (b) Yes. 101 − 1 = 100 = 4 × 25, so 101 is the 25th term. (c) No. Every term is 1 more than a multiple of 4, so every term is odd; 150 is even. (150 − 1 = 149 is not a multiple of 4.)

Q4 Trace: x is 7, 12, 17, 22. (a) 22 (b) 502. x starts at 2 and 100 jumps of 5 add 500.

Q5 Program: when green flag clicked; set n to 4; repeat 20 [say n for 1 seconds; change n by 6]. Say must come before change (or start n at −2 and change first). (a) 118 (the 20th term: 6 × 20 − 2), the last number it says. It says one term per second.

Q6 (a) 203 (3 + 50 × 4) (b) 199 (4 × 50 − 1) (c) Change repeat 50 to repeat 49. The 1st term is already there before the loop starts, so the 50th term is only 49 jumps away.

Investigation Regular shapes with k sides joined along one side each time use (k − 1) × n + 1 sticks: triangles 2n + 1, squares 3n + 1, pentagons 4n + 1, hexagons 5n + 1. What changes: the number added each time, which is always one less than the number of sides because a side is shared. What stays the same: the rule has the form (something) × n + 1, and pattern 1 is the single shape (k sticks). Look for: a table, the difference named, a check of the rule on pattern 1 and one more pattern, and a reason for the sharing. Any shape chosen is valid. Stretch: a child may write the rule for any k, or test it in code (set s to k, repeat n − 1 [change s by k − 1]).

Lesson 2: Prime Hunters

Week 1 (second lesson) • Primes, Factors and Prime Factorisation • Workbook: Lesson 2

Objective	Find all the factors of a number using factor pairs, decide whether a number is prime by testing small divisors, write a number as a product of primes with a factor tree, use mod (the remainder), if and a counter in code to count factors, and explore Goldbach pairs and numbers with exactly 3 factors.
Thinking skills	Systematic listing (factor pairs), testing and ruling out, reasoning about when to stop, tracing code, generalising (squares of primes)
Materials	Laptop or desktop with Scratch 3 open, a printed 1 to 100 grid (or squared paper), two coloured pencils, mini whiteboard, workbook. Optional: a Python editor for children who already know Python.
Online set-up	Share the 1 to 100 grid on screen and annotate it while children work on their own printed or hand-drawn grid. Build the prime checker together in your shared Scratch window; the blocks are also on the slide. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. Children hold their grid up to the camera at minute 22.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Welcome back, inquirers! Today we hunt primes, the building blocks of every whole number."*

"People have hunted primes for over 2000 years, and today computers hunt them to keep the internet safe. We hunt by hand first, then teach a computer to do it."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

2–10 min Warm-up

DO Warm-up check first (slide 2, 3 minutes). It comes out of the 10-minute warm-up: play the game for about 3 minutes and ask only the first three quick questions after it.

SAY *"Two quick puzzles. They are new, not from last time. Write your answer AND one reason on your whiteboard. Hold it up when I say."*

Warm-up check	Answer and the reason to listen for
A tutor wants to split 31 children into equal teams, with more than 1 child in each team and more than 1 team. Is that possible? Why?	No. 31 is prime: its only factors are 1 and 31, so the only equal groups are 1 team of 31 or 31 teams of 1.
Which has more factors, 36 or 37? How do you know?	36 has 9 factors (1, 2, 3, 4, 6, 9, 12, 18, 36). 37 is prime and has only 2 (1 and 37). A bigger number does not mean more factors.

DO Fast track (nobody is labelled). If a child has BOTH answers right and a sensible reason, send a private chat message: “Today start on the Think Harder page at minute 40.” Fast-track children still join Explore, the mini-lesson and Talk time; only the worksheet time changes. Tell everyone: “Some of you start on a different page today. It changes every lesson, and you can move any time.” Never say “advanced”, never record the check as a mark, and let any child choose the main pages. A fast-track child who gets stuck goes back to the main page. This works because groups are small (up to 8) and each child has a printed workbook, so you can give children different pages quietly.

DO Factor pair race. Six number cards. Pairs race to say every factor pair. One card has only one pair: which one?

SAY *“Factor pair race. I show a number. With your partner, find every pair of whole numbers that multiply to make it. First pair to say them all wins the card.”*

“One card is special. See if you can spot which one.”

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
17 ÷ 5 = 3 remainder __	2
30 ÷ 6 = 5 remainder __	0
23 ÷ 4 = 5 remainder __	3
100 ÷ 7: what is the remainder?	2 (7 × 14 = 98)
A remainder of 0 means ...?	It divides exactly: it is a factor.

10–25 min Explore (in pairs, with objects)

SAY *“How many prime numbers do you think there are from 1 to 100? Write a guess. Then we find out with a 2000-year-old method called the sieve.”*

DO 1. On the 1 to 100 grid, cross out 1. It is not prime.

2. Circle 2. Cross out every other multiple of 2.

3. Circle the next number not crossed out (3) and cross out its multiples. Do the same for 5 and 7.

4. Circle every number left. Count the primes.

SAY *“Talk to your partner while you work. I will come round and ask you questions.”*

IF STUCK Stuck: circle 2, then cross out 4, 6, 8, ... every 2nd number. Then circle 3 and cross out every 3rd number.

Crossing out the same number twice: “That is fine. 6 is a multiple of 2 AND of 3.”

Finished early: circle 11 and look for a multiple of 11 below 100 that is not crossed out yet. Is there one? Why not?

DO At about minute 22, say “Pencils down” and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How many primes are there from 1 to 100?	25.
Why could we stop after crossing out the multiples of 7?	The next prime is 11, and 11 × 11 = 121 is bigger than 100. Every smaller multiple of 11 (22, 33, … 99) has a smaller factor, so it is already crossed out.
Is 1 prime? Is 2 prime?	1 is not: it has only one factor. 2 is prime (factors 1 and 2), the only even prime.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *“Factors of 36. I work in pairs: 1 × 36, 2 × 18, 3 × 12, 4 × 9, 6 × 6. The next would be 9 × 4, which I already have, so I stop. 9 factors, because 6 only counts once.”*

“Is 91 prime? It is not even. 9 + 1 = 10, not a multiple of 3. It does not end in 0 or 5. Try 7: 7 × 13 = 91. So 91 is NOT prime. It looks prime, but it is not.”

“How far do I need to test? 10 × 10 = 100 is bigger than 91, so if 91 had a factor pair, one of the pair would be less than 10. I only test 2, 3, 5 and 7.”

“In code, n mod d is the remainder when n is divided by d. If it is 0, d is a factor, so I change count by 1. At the end, count = 2 means prime.”

Worked example on the slides	Answer
Is 91 prime?. Not even; 9 + 1 = 10 is not a multiple of 3; does not end in 0 or 5. Try 7: 91 ÷ 7 = 13. Stop testing once d × d is bigger than the number (10 × 10 = 100).	Not prime: 91 = 7 × 13.
Prime checker in code (n = 15). repeat n: if n mod d = 0 then change count by 1; change d by 1. n mod d is the remainder. 0 means d is a factor. At the end, count = 2 means prime.	Factors 1, 3, 5 and 15: count = 4, not 2, so 15 is not prime.

SAY *“The key idea today is: exactly two factors. A prime has exactly 2 factors: 1 and itself. Test 2, 3, 5, 7, … until d × d is bigger than the number. Say it with me.”*

DO Show the key idea slide at minutes 33 to 35. The “common mistakes” on that slide are for you, not the children. Watch for them during the worksheet:

- Saying 1 is prime, or that 2 is not prime because it is even.

- Missing factors by not working in pairs, or counting the middle factor of a square number twice (6 in 36).
- Stopping a factor tree at a number that is not prime (72 = 2 × 2 × 18), or reading mod as division (thinking 10 mod 4 is 2.5 instead of the remainder 2).

35–40 min Brain break

DO Prime or not? Jump for prime, squat for not prime: 7, 9, 13, 21, 2, 1, 31, 27, 51, 53.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

40–75 min Worksheet (pairs, then alone, then stretch)

DO Minute 40: give out the workbook. Read the title of Lesson 2 aloud. Say "Pairs first. Talk before you write."

Minutes 40 to 55: pairs work together on the first page (with the short teaching slides). Minutes 55 to 72: each child works alone on the second page. Minutes 72 to 75: check together, then stretch and challenge for early finishers.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "How do you know you have found every factor?"
- Ask: "Why can you stop testing at 7 for numbers below 100?"
- Ask: "Point to the block that adds 1 to count. When does it run?"

IF STUCK Hint 1: "Start with 1 × the number, then try 2, 3, 4 … and stop when the pairs meet."

Hint 2: "Use the quick tests: even? digits add to a multiple of 3? ends in 0 or 5? Then try 7."

Hint 3: "Split the number into any two factors you know, then keep splitting until every branch ends in a prime."

DO Early finishers: give the stretch question first, then the challenge, then the Open investigation (it is the last box on the second Think Harder page). Fast-track children start there at minute 40.

Stretch and challenge questions

Question	Answer
Workbook Think Harder 1: How many factors does 64 have? List them.	7: 1, 2, 4, 8, 16, 32, 64 (8 × 8 counts once).
Workbook Think Harder 2: Find the smallest number that has exactly 6 factors.	12 (1, 2, 3, 4, 6, 12). Every number below 12 has fewer than 6 factors.
Workbook Think Harder 3: Write 100 as the sum of two primes in every possible way.	6 ways: 3 + 97, 11 + 89, 17 + 83, 29 + 71, 41 + 59, 47 + 53.

Workbook Think Harder 4: What does this code say? set count to 0, set d to 1, then repeat 45 [if 45 mod d = 0 then change count by 1; change d by 1], then say count.	6. It counts the factors of 45: 1, 3, 5, 9, 15, 45.
Workbook Challenge 5: Make up a “Who am I?” number puzzle using factors and primes, for example: “I am between 30 and 60 and I have exactly three factors.” Write 2 or 3 clues so that only ONE number fits. Swap with a partner, then check that no other number fits.	Open. Example: “I am between 30 and 60 and I have exactly 3 factors” fits only 49 (7 × 7). The numbers with exactly 3 factors are the squares of primes: 4, 9, 25, 49, 121, … Check a puzzle by listing every number that fits the clues.
Workbook Challenge 6: A farmer counts her eggs in 2s, 3s, 4s, 5s and 6s, and each time there is 1 egg left over. When she counts them in 7s, none are left over. What is the smallest number of eggs she could have?	301. It must be 1 more than a multiple of 60: 61, 121, 181, 241, 301, … The first of these in the 7 times table is 301 = 7 × 43.
Workbook Challenge 7: 20 lockers are closed. Student 1 opens every locker. Student 2 changes every 2nd locker (open to closed, closed to open). Student 3 changes every 3rd locker, and so on up to student 20. Which lockers are open at the end?	1, 4, 9 and 16. Locker n is changed once for each factor of n. Only square numbers have an odd number of factors, so only they end up open.
Workbook Open investigation (Is it always true? Test and explain): Ali says: “If you add two different primes, the answer is never a prime.” Test him. List every pair of different primes below 20 and add them. Is Ali right? What do you notice about the pairs that do make a prime? Explain why.	Ali is wrong. The primes below 20 are 2, 3, 5, 7, 11, 13, 17, 19, giving 28 pairs. Exactly four sums are prime: 2 + 3 = 5, 2 + 5 = 7, 2 + 11 = 13, 2 + 17 = 19. Every one of them uses the prime 2. Why: two odd primes add to an even number bigger than 2, which has 2 as a factor, so it is not prime. 2 is the only even prime, so only a pair with 2 can give an odd sum, and then the sum is prime only if the other prime plus 2 is also prime (3, 5, 11, 17 work; 7, 13, 19 do not). Look for: a systematic list of all 28 pairs, the spot of the number 2, a reason using odd and even. One counter-example is enough to show “never” is false.

These are the questions on the two Think Harder pages in the workbook. More are in the separate Question Bank.

75–82 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask two or three children to share, not only the fastest.

SAY *“Why does a square number like 36 have an odd number of factors?”*

“Is there a biggest prime? What do you think?”

Discussion question	Listen for

Why does a square number like 36 have an odd number of factors, when most numbers have an even number?	Factors come in pairs that multiply to the number. For a square number one pair is a number times itself (6 × 6), and that factor is counted only once, so the total is odd. There is no biggest prime: Euclid showed this about 2300 years ago (a teaser for secondary school).

82–90 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
(a) Is 133 prime? Show your test. (b) Write 66 as a product of primes.	(a) No. Not even, digits add to 7, does not end in 0 or 5, but 133 ÷ 7 = 19, so 133 = 7 × 19. (b) 66 = 2 × 3 × 11

SAY *"You sieved, you tested, you split numbers into primes, and you taught a computer to count factors. Real prime hunters!"*

"Do the exit ticket on your own, then show me."

Reflection: What was hard? (1 minute)

SAY *"Last minute, just for you. On the last page of today's lesson, finish the three sentences in the box called "What was hard?": the hardest part was, I got unstuck by, next time I will try. Nobody marks it, and nobody has to show it. Then I would like to hear ONE sentence from two or three of you. "Pass" is fine."*

DO Run it at about minute 88, while you collect exit tickets (slide at the end). Never grade or correct it, and never read it out without the child's say-so. Praise the strategy, not the speed: say "you you listed the factors in pairs — that is what got you unstuck", never "you were quick". If a child writes "nothing was hard", ask "what would make it harder?".

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Extension for children who are ready

First add if count = 2 then say prime, else say not prime after the loop. Then put the prime checker inside a second loop so n goes 1, 2, 3, … 100 and the sprite says only the primes. Does it find 25 primes, like the sieve?

Answer key (workbook)

Q1 28: 1, 2, 4, 7, 14, 28 (6 factors, not prime). 31: 1, 31 (2, prime). 49: 1, 7, 49 (3, not prime). 51: 1, 3, 17, 51 (4, not prime: 51 = 3 × 17). 1: 1 (1 factor, not prime). (a) 67, 83 and 97 are prime (57 = 3 × 19, 77 = 7 × 11, 99 = 9 × 11). (b) No. 8 + 7 = 15, a multiple of 3, so 87 = 3 × 29.

Q2 (a) 72 = 2 × 2 × 2 × 3 × 3 (b) 150 = 2 × 3 × 5 × 5 (c) 126 (2 × 3 × 3 × 7)

Q3 d = 1 to 6. 6 mod d: 0, 0, 0, 2, 1, 0. count after each loop: 1, 2, 3, 3, 3, 4. (a) 4 (the factors 1, 2, 3, 6) (b) It says 2, so 13 is prime: its only factors are 1 and 13.

Q4 (a) 24 = 5 + 19 = 7 + 17 = 11 + 13 (3 ways) (b) 36 = 5 + 31 = 7 + 29 = 13 + 23 = 17 + 19 (4 ways) (c) An odd total needs one even and one odd number. The only even prime is 2, and 27 − 2 = 25 is not prime.

Q5 (a) 4, 9, 25, 49 (b) They are squares of primes (2 × 2, 3 × 3, 5 × 5, 7 × 7). Their factors are 1, the prime and the square. (c) 121 (11 × 11)

Q6 E, C, B. Inside B: F (with A inside F), then D. It says 1, 2, 4, 5, 8, 10, 20, 40 (8 numbers, one per second).

Investigation Ali is wrong. The primes below 20 are 2, 3, 5, 7, 11, 13, 17, 19, giving 28 pairs. Exactly four sums are prime: 2 + 3 = 5, 2 + 5 = 7, 2 + 11 = 13, 2 + 17 = 19. Every one of them uses the prime 2. Why: two odd primes add to an even number bigger than 2, which has 2 as a factor, so it is not prime. 2 is the only even prime, so only a pair with 2 can give an odd sum, and then the sum is prime only if the other prime plus 2 is also prime (3, 5, 11, 17 work; 7, 13, 19 do not). Look for: a systematic list of all 28 pairs, the spot of the number 2, a reason using odd and even. One counter-example is enough to show “never” is false.

Lesson 3: Below Zero

Week 2 (first lesson) • Negative Numbers • Workbook: Lesson 3

Objective	Order and compare negative numbers on a number line; add and subtract with negative numbers by moving along the line, in temperature, lift and sea-level contexts; use "change by" a negative number in code and trace a variable that goes below 0.
Thinking skills	Number line reasoning, splitting a jump at 0, difference as a distance, tracing code, debugging, planning a loop
Materials	Laptop or desktop with Scratch 3 (scratch.mit.edu, free, no log-in needed), mini whiteboard and marker, workbook, pencil.
Online set-up	Children keep Scratch open in one browser tab and the video call in another. The tutor screen-shares Scratch to show the x number under the stage. Children share their screens only when asked; page 1 of the worksheet is done on paper with laptops turned away. Type the minus sign with the keyboard minus key (the hyphen); do not copy and paste a number from the slide or chat, because the slide uses a different minus character that Scratch will not accept. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Welcome back, inquirers! In Secondary 1 you will meet numbers that go below zero. You already know some: the freezer, a winter holiday in Japan, the basement car park. Today we find out how they behave, and we make the computer go below zero too."*

"Notice it, test it, explain why. Today "test it" means test it in Scratch."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

2–10 min Warm-up

DO Warm-up check first (slide 2, 3 minutes). It comes out of the 10-minute warm-up: play the game for about 3 minutes and ask only the first three quick questions after it.

SAY *"Two quick puzzles. They are new, not from last time. Write your answer AND one reason on your whiteboard. Hold it up when I say."*

Warm-up check	Answer and the reason to listen for
A freezer is at −18 °C and a fridge is at 4 °C. How many degrees warmer is the fridge than the freezer?	22 degrees. From −18 up to 0 is 18, then 4 more: 18 + 4 = 22. A number line drawing shows it.
A sprite is at x = −40. It runs "change x by 25", then "change x by −60". Where does it end up?	x = −75. −40 + 25 = −15, then −15 − 60 = −75. The sprite is left of zero, and further left than where it started.

DO Fast track (nobody is labelled). If a child has BOTH answers right and a sensible reason, send a private chat message: “Today start on the Think Harder page at minute 40.” Fast-track children still join Explore, the mini-lesson and Talk time; only the worksheet time changes. Tell everyone: “Some of you start on a different page today. It changes every lesson, and you can move any time.” Never say “advanced”, never record the check as a mark, and let any child choose the main pages. A fast-track child who gets stuck goes back to the main page. This works because groups are small (up to 8) and each child has a printed workbook, so you can give children different pages quietly.

DO Cold city line-up. Five cities at 7 a.m. on a December morning: Singapore 25 °C, Tokyo 3 °C, Seoul −7 °C, Beijing −9 °C, Harbin −22 °C. Put them in order from coldest to warmest, then find how many degrees warmer Tokyo is than Seoul.

SAY *“Here are five cities at 7 a.m. on a December morning. Put them in order from coldest to warmest. Beijing is −9 and Seoul is −7. Nine is bigger than seven, so which is colder? Argue with your partner.”*

“How many degrees warmer is Tokyo than Seoul? How did you count it?”

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
It is 3 °C. It falls 5 degrees. What is it now?	−2 °C
Which is colder: −7 °C or −2 °C?	−7 °C
Count back in 3s: 6, 3, 0, __, __	−3, −6
What number is 4 less than 1?	−3
A lift is at B1. It goes down 2 floors. Where is it?	B3

10–25 min Explore (in pairs, with objects)

SAY *“Your sprite is at x = 0. It changes x by 100, then changes x by −250. Where will it be? Write your prediction before you test it.”*

“The thermometer starts at 5 and goes down 3, six times. What is the last number it will say? Write it down first.”

DO 1. In Scratch, set x to 0: the sprite is in the middle of the stage.

2. Predict, then test: change x by 100, then change x by −250. Where is the sprite?

3. Build the thermometer: set temp to 5, repeat 6: change temp by −3, say temp.

4. Predict the last number it says. Then click the flag and check.

SAY *“Talk to your partner while you work. I will come round and ask you questions.”*

IF STUCK Stuck: drag the sprite left and right and watch the x number under the stage. Left of the middle is negative.

Wrong prediction: draw a number line on your whiteboard and do the six jumps of 3. Where do you cross 0?

Finished early: make the sprite walk with the temperature (set x to temp × 10), or add "if temp < 0 then say Below zero!".

DO At about minute 22, say “Pencils down” and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
Where did the sprite end up after +100 and then −250?	x = −150
What was the last number the thermometer said?	−13 (2, −1, −4, −7, −10, −13)
How do we make a variable go down in Scratch?	Change it by a negative number: adding a negative moves left.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *“It is −4 °C and it rises 9 degrees. Rising means moving right. From −4 it takes 4 to get to 0. I still have 5 of my 9 left, so I land on 5 °C.”*

“Change temp by −4 means I add negative 4. Adding a negative moves me left, so it is the same as taking away 4: 1 + (−4) = 1 − 4 = −3.”

“7 − (−3) asks how far it is from −3 to 7. From −3 up to 0 is 3, from 0 up to 7 is 7. That is 10 steps right, so the answer is 10. Check: −3 + 10 = 7.”

Worked example on the slides	Answer
Up and down the line. It is −4 °C at 6 a.m. It rises 9 degrees by noon, then falls 7 degrees by night. Split each jump at 0: −4 up to 0 is 4, then 5 more; 5 down to 0, then 2 more. What is the temperature at noon, and at night?	Noon 5 °C, night −2 °C.
Subtracting: the jump from b to a. How much warmer is 7 °C than −3 °C? That is 7 − (−3). Start at −3 and count to 7: 3 up to 0, then 7 more. What is 7 − (−3)?	10 (check: −3 + 10 = 7).

SAY *“The key idea today is: move along the number line. Right is bigger. Adding a positive moves right; adding a negative moves left. a − b is the jump from b to a. Say it with me.”*

DO Show the key idea slide at minutes 33 to 35. The “common mistakes” on that slide are for you, not the children. Watch for them during the worksheet:

- Ordering negatives by their digits only (saying −9 is bigger than −4).
- Crossing zero wrongly, e.g. 3 − 8 answered as 5 or −11, instead of splitting the jump at 0 (3 down to 0, then 5 more: −5).
- In code, using "change by 1" when the variable should go down, or counting the start value as one of the repeats when tracing.

35–40 min Brain break

DO Human lift. Standing tall is floor 0; reach up for each floor up, bend lower for each floor down. Moves: up 3, down 5, down 2, up 6. Call out the floor after each move (3, −2, −4, 2).

Water. Toilet break for anyone who needs it. Keep it to five minutes.

40–75 min Worksheet (pairs, then alone, then stretch)

DO Minute 40: give out the workbook. Read the title of Lesson 3 aloud. Say “Pairs first. Talk before you write.”

Minutes 40 to 55: pairs work together on the first page (with the short teaching slides). Minutes 55 to 72: each child works alone on the second page. Minutes 72 to 75: check together, then stretch and challenge for early finishers.

SAY *“Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking.”*

DO Circulate and ask:

- Ask: “Which way did you move? Where did you cross zero?”
- Ask: “How can you check your answer by adding back?”
- Ask: “In your trace table, which row is the start and which rows are the repeats?”

IF STUCK Hint 1: “Draw a number line and mark 0 first.”

Hint 2: “Split the jump at 0: how far to 0, and how far after it?”

Hint 3: “For a − b, start at b and count to a. Right means a positive answer, left means a negative answer.”

DO Early finishers: give the stretch question first, then the challenge, then the Open investigation (it is the last box on the second Think Harder page). Fast-track children start there at minute 40.

Stretch and challenge questions

Question	Answer
Workbook Think Harder 1: Work out −7 + 12 and 5 − 13.	5 and −8
Workbook Think Harder 2: A diver is at −14 m. A seagull is flying at 9 m. How far apart are they?	23 m (14 up to sea level, then 9 more)
Workbook Think Harder 3: Work out 4 − (−6). Hint: how far is it from −6 to 4?	10

Workbook Think Harder 4: Find the missing number: −3 + ____ = −10.	−7 (adding −7 moves 7 to the left)
Workbook Challenge 5: Make up your own temperature story with at least three changes (rises and falls) that starts below zero and ends at exactly −7 °C. Write the story, the working and a number-line sketch. Swap with a partner and check each other's.	Open. Example: starts at −2 °C, falls 6 degrees (−8), rises 3 degrees (−5), falls 2 degrees (−7). Check that the story really ends at −7 and that each change is shown on a number line.
Workbook Challenge 6: Two numbers add up to −2. Their difference is 10. What are the two numbers?	4 and −6 (4 + (−6) = −2 and 4 − (−6) = 10)
Workbook Challenge 7: In many Singapore buildings there is no level 0: level 1 is the ground floor and B1 is the level below it. A lift goes from level 4 down to B3. How many floors does it go down? (Count the levels one by one; do not use a number line with 0.)	6 floors (4, 3, 2, 1, B1, B2, B3: six moves down). A number line with 0 would give 7, because it counts a level 0 that these buildings do not have.
Workbook Open investigation (Find all the ways): Here are some numbers: −5, −3, −1, 0, 2, 4, 6. Find ALL the pairs (a, b) from the list with a − b = 7. Mark each pair on a number line. What do you notice? Then try again with a − b = 3.	a − b = 7: (2, −5), (4, −3) and (6, −1): exactly three pairs (checked by trying every pair in the list). On the number line each pair is the same distance, 7, apart: the same jump slid along the line. a − b = 3: (0, −3) and (2, −1): two pairs. What stays the same: the gap between a and b. Look for: a systematic search (take each b and ask what a would be, then check a is on the list), a jump drawn on a number line, and the idea that "difference" means distance. Children who say (7, 0) have gone outside the list: ask "is 7 on the list?".

These are the questions on the two Think Harder pages in the workbook. More are in the separate Question Bank.

75–82 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask two or three children to share, not only the fastest.

SAY *"Ben says −8 is bigger than −3 because 8 is bigger than 3. Is he right?"*

"Why does change by −3 make a number go down?"

Discussion question	Listen for
Ben says −8 is bigger than −3 because 8 is bigger than 3. Is he right? Use a thermometer or a lift to explain.	Ben is not right. −8 is further left on the number line, so it is smaller: −8 °C is colder than −3 °C and floor −8 is lower than floor −3. The digit tells us how far from 0, the sign tells us which side. Change by −3 adds a negative, which moves left, so it is the same as taking away 3.

82–90 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
It is −3 °C. The temperature falls 6 degrees, then rises 4 degrees. What is the temperature now?	−5 °C (−3 − 6 = −9, then −9 + 4 = −5)

SAY *"Today you went below zero on paper and in code. In Secondary 1 these are called negative numbers or integers, and you already know how to move with them. Well done, inquirers!"*

Reflection: What was hard? (1 minute)

SAY *"Last minute, just for you. On the last page of today's lesson, finish the three sentences in the box called "What was hard?": the hardest part was, I got unstuck by, next time I will try. Nobody marks it, and nobody has to show it. Then I would like to hear ONE sentence from two or three of you. "Pass" is fine."*

DO Run it at about minute 88, while you collect exit tickets (slide at the end). Never grade or correct it, and never read it out without the child's say-so. Praise the strategy, not the speed: say "you you drew a number line — that is what got you unstuck", never "you were quick". If a child writes "nothing was hard", ask "what would make it harder?".

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Extension for children who are ready

Make Dot walk along the stage as the temperature changes: inside the loop add "set x to temp × 10" and "if temp < 0 then say Below zero!". Predict where Dot stops before running it.

Answer key (workbook)

Q1 (a) A = −8, B = −3, C = 4 (b) −12, −5, −1, 0, 3, 7 (c) −4 > −9; −1 < 0; −15 < 2

Q2 (a) 3 p.m. (−11 °C) (b) 11 degrees (−24 up to −13) (c) 9 degrees (−11 down to −20) (d) −25 °C (e) 44 degrees (13 up to 0, then 31 more)

Q3 (a) 15 m (6 up to sea level, then 9 more) (b) 9 m (−6 down to −15) (c) −13 m (−6 + 4 = −2, then −2 − 11 = −13)

Q4 (a) 3 (b) −7 (c) −8 (d) −8 (e) 8 (from −3 to 5) (f) 6 (from −7 to −1)

Q5 (a) Table: start 40; 1: 10; 2: −20; 3: −50; 4: −80. The sprite says −80. (b) 4 more repeats (−80 to −200 is 120, and 120 ÷ 30 = 4)

Q6 (a) 13 (the lift goes up: 5 + 8) (b) change floor by −1 (c) −3, the third basement (B3)

Q7 (a) set depth to 0; repeat 7; change depth by −4; say depth (7 × 4 = 28) (b) 28 (change depth by 28 brings the diver back to 0 m)

Investigation a − b = 7: (2, −5), (4, −3) and (6, −1): exactly three pairs (checked by trying every pair in the list). On the number line each pair is the same distance, 7, apart: the same jump slid along the line. a

– b = 3: (0, –3) and (2, –1): two pairs. What stays the same: the gap between a and b. Look for: a systematic search (take each b and ask what a would be, then check a is on the list), a jump drawn on a number line, and the idea that “difference” means distance. Children who say (7, 0) have gone outside the list: ask “is 7 on the list?”.

Lesson 4: Algebra Machines

Week 2 (second lesson) • Expressions and Equations • Workbook: Lesson 4

Objective	Find the rule of a function machine from a table and write it with a letter (OUT = 3n + 2); see that the order of the steps matters and use brackets; solve simple equations by running the machine backwards and checking forwards; build an input-rule-output program and use a search loop to test an answer.
Thinking skills	Rule finding from a table, writing expressions with a letter, order of operations and brackets, inverse operations, solving simple equations, tracing loops, explaining a number trick with algebra
Materials	Laptop or desktop with Scratch 3 (scratch.mit.edu, free), mini whiteboard and marker, workbook, pencil.
Online set-up	Pairs work in breakout rooms for the explore: one partner shares their screen with the machine running and types in the numbers the other partner calls out. Page 1 of the worksheet is done on paper with laptops turned away. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Today you are machine builders and machine breakers. Every machine has a secret rule. In Secondary 1 the rule is written with a letter, like 3n + 2, and that is all algebra is: a machine written down."*

"Notice it, test it, explain why. Today we test our rules in code."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

2–10 min Warm-up

DO Warm-up check first (slide 2, 3 minutes). It comes out of the 10-minute warm-up: play the game for about 3 minutes and ask only the first three quick questions after it.

SAY *"Two quick puzzles. They are new, not from last time. Write your answer AND one reason on your whiteboard. Hold it up when I say."*

Warm-up check	Answer and the reason to listen for
A phone plan costs $10 a month plus $3 for every GB of data. Write the cost for n GB, then find the cost for 7 GB.	Cost = 3n + 10. For 7 GB: 3 × 7 + 10 = $31.
One month the same plan cost $52. How many GB were used?	14 GB. Undo the + 10 first: 52 − 10 = 42. Then undo the × 3: 42 ÷ 3 = 14. Check: 3 × 14 + 10 = 52.

DO Fast track (nobody is labelled). If a child has BOTH answers right and a sensible reason, send a private chat message: “Today start on the Think Harder page at minute 40.” Fast-track children still join Explore, the mini-lesson and Talk time; only the worksheet time changes. Tell everyone: “Some of you start on a different page today. It changes every lesson, and you can move any time.” Never say “advanced”, never record the check as a mark, and let any child choose the main pages. A fast-track child who gets stuck goes back to the main page. This works because groups are small (up to 8) and each child has a printed workbook, so you can give children different pages quietly.

DO Guess my rule. The tutor is a machine with a secret rule. Call out IN numbers, hear the OUT, and guess the rule. Rounds: n + 5, 5n, 2n − 1, 3n + 2. Check each rule on every pair.

SAY *“I am a machine. Call out a number and I will tell you what comes out. You say 3, I say 8. You say 7, I say 12. What is my rule?”*

“Which numbers did you choose to call out? Which ones helped you most?”

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
n = 6. What is 4n?	24
n = 6. What is n + 9?	15
n = 6. What is 3n − 5?	13
2n = 18. What is n?	9
n = 6. What is (n + 4) × 2?	20

10–25 min Explore (in pairs, with objects)

SAY *“My machine does times 3, then add 2. If you type 4, what will it say? Write your prediction before we run the code.”*

“If I swap the steps, add 2 first and then times 3, will 4 give the same answer? Thumbs up or down.”

DO 1. Build the machine in Scratch: ask IN? and wait, set out to (answer * 3) + 2, say out.

2. Change the rule to your own secret two-step rule.

3. Share your screen. Your partner calls IN numbers and you type them in.

4. Your partner writes your rule with n, using as few IN numbers as possible. Then swap.

SAY *“Talk to your partner while you work. I will come round and ask you questions.”*

IF STUCK Stuck building: put the * block inside the + block, then put “answer” in the first gap of the *.

Stuck cracking a rule: ask for IN 0, 1, 2, 3. How much does OUT go up each time?

Finished early: make a three-step machine, or put ask and say in a forever loop.

DO At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
Which IN numbers were most useful?	Numbers in a row show the × part; IN 0 shows the + part.
How did you write the rule with n?	The × part first, then the + or − part, e.g. 3n + 2.
The machine × 3, + 2 says 20. How can you find IN without guessing?	Run it backwards: 20 − 2 = 18, 18 ÷ 3 = 6.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"IN goes up by 1 and OUT goes up by 4 each time. That is what times 4 does. 4 × 1 is 4 but OUT is 7, so I add 3. Rule: 4n + 3. Check: 4 × 4 + 3 = 19."*

"Machine A adds 2 then times 5; machine B times 5 then adds 2. For 4: (4 + 2) × 5 = 30 and 4 × 5 + 2 = 22. Same steps, different order, different answer. The brackets show what happens first."

"5n − 4 = 31. The machine did times 5, then take away 4. I undo the last step first: 31 + 4 = 35. Then undo times 5: 35 ÷ 5 = 7. Check: 7 × 5 − 4 = 31."

Worked example on the slides	Answer
From a table to a rule. IN 1, 2, 3, 4 gives OUT 7, 11, 15, 19. OUT goes up 4 each time, so × 4. 4 × 1 = 4, but OUT is 7, so + 3. What is the rule? Check it on IN 4.	OUT = 4n + 3 (4 × 4 + 3 = 19).
Run the machine backwards. A machine does × 5, then − 4. OUT is 31. That is 5n − 4 = 31. Undo the last step first: 31 + 4 = 35. Then undo × 5: 35 ÷ 5. What is n? Check forwards.	n = 7 (7 × 5 − 4 = 31).

SAY *"The key idea today is: a rule is a machine. OUT = 3n + 2 means × 3, then + 2. To find IN from OUT, run the machine backwards: undo the last step first. Say it with me."*

DO Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Finding a rule from the first pair only (1 → 6 "add 5") without checking it on the other pairs.
- Undoing the steps in the wrong order, or doing the same step again instead of its opposite.
- Ignoring brackets or the order of operations: reading 4n + 1 as "add 1, then times 4".

35–40 min Brain break

DO Jumping machine. Rule: double it, then take away 1. The tutor calls IN 2, 3, 4, 5; everyone does that many OUT star jumps (3, 5, 7, 9). Last round: "I did 11 jumps. What was my IN?" (6).

Water. Toilet break for anyone who needs it. Keep it to five minutes.

40–75 min Worksheet (pairs, then alone, then stretch)

DO Minute 40: give out the workbook. Read the title of Lesson 4 aloud. Say "Pairs first. Talk before you write."

Minutes 40 to 55: pairs work together on the first page (with the short teaching slides). Minutes 55 to 72: each child works alone on the second page. Minutes 72 to 75: check together, then stretch and challenge for early finishers.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "Did you check your rule on every pair, not just the first?"
- Ask: "What was the last step? What is its opposite?"
- Ask: "How could the computer check your answer?"

IF STUCK Hint 1: "How much does OUT go up when IN goes up by 1? That is the × part."

Hint 2: "Draw the machine boxes. Write the OUT at the end and go backwards."

Hint 3: "Put your answer back into the machine. Do you get the OUT?"

DO Early finishers: give the stretch question first, then the challenge, then the Open investigation (it is the last box on the second Think Harder page). Fast-track children start there at minute 40.

Stretch and challenge questions

Question	Answer
Workbook Think Harder 1: IN 1, 2, 3, 4 gives OUT 1, 7, 13, 19. Find the rule. What is OUT when IN is 20?	OUT = 6n − 5; IN 20 gives 115.
Workbook Think Harder 2: Solve 9n − 7 = 65.	n = 8 (65 + 7 = 72, 72 ÷ 9 = 8)
Workbook Think Harder 3: Machine A does × 2, then + 10. Machine B does + 5, then × 2. Put 7 into both. What do you notice? Explain with n.	Both give 24. (n + 5) × 2 = 2n + 10, so they always give the same OUT.
Workbook Think Harder 4: I think of a number, divide it by 4, then add 6. I get 15. What is my number?	36 (15 − 6 = 9, 9 × 4 = 36)
Workbook Challenge 5: Find two different machines of the form "× a number, then + or − a number" that both turn IN 5 into OUT 18. Then find the machine of this form that ALSO turns IN 2 into OUT 6. Is there only one?	Many machines turn 5 into 18: × 3 then + 3, × 2 then + 8, × 1 then + 13, × 4 then − 2, … Only × 4 then − 2 also turns 2 into 6 (4 × 2 − 2 = 6). Two pairs of IN and OUT fix the rule: 5a + b = 18 and 2a + b = 6 give 3a = 12, so a = 4 and b = −2.

Workbook Challenge 6: Mei is n years old. Her mother is 3 times as old. In 12 years, her mother will be twice as old as Mei. How old is Mei now?	12 (3n + 12 = 2(n + 12), so n = 12; mother 36; in 12 years 24 and 48)
Workbook Challenge 7: Ali's secret machine does × a number, then + or − a number. IN 4 gives OUT 9. IN 9 gives OUT 24. What does IN 20 give? Which IN gives OUT 99?	The rule is 3n − 3. IN 20 gives 57. OUT 99 comes from IN 34.
Workbook Open investigation (Find all the ways): A machine does "× a number, then + a number". Use whole numbers: the × number is from 1 to 9 and the + number is from 0 to 9. Find ALL the machines that turn IN 4 into OUT 17. What pattern do you spot in your answers? Then try IN 4 and OUT 21.	IN 4, OUT 17: × 2 then + 9, × 3 then + 5, × 4 then + 1. (× 1 would need + 13, too big; × 5 would need −3, not allowed.) That is exactly three machines. Pattern: when the × number goes up by 1, the + number goes down by 4, because IN is 4 and the total must stay 17. IN 4, OUT 21: × 3 then + 9, × 4 then + 5, × 5 then + 1: again three machines, again down by 4. Look for: a table of × number and + number, a systematic order (try × 1, × 2, × 3, ...), and the explanation of the "down by 4". Stretch: a different IN gives a different step (IN 5 steps by 5).

These are the questions on the two Think Harder pages in the workbook. More are in the separate Question Bank.

75–82 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask two or three children to share, not only the fastest.

SAY *"To solve 4n + 7 = 43, Ravi tries n = 1, 2, 3 ... like the computer. Mei runs the machine backwards. Which way is better, and when?"*

"Why do we undo the LAST step first?"

Discussion question	Listen for
To solve 4n + 7 = 43, Ravi tries n = 1, 2, 3 ... like the computer. Mei runs the machine backwards. Which way is better, and when?	Both give n = 9. Running backwards is quicker by hand and works for big numbers. Trying every n is easy to code and also works when there is no easy backwards step (like n × n + n = 90), but it is slow by hand and must stop if there is no whole-number answer. We undo the last step first because it was the last thing done, like taking off shoes before socks.

82–90 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
A machine does × 4, then + 5. (a) IN 6: what is OUT? (b) OUT 45: what was IN?	(a) 29 (b) 10 (45 − 5 = 40, 40 ÷ 4 = 10)

SAY *"Today you wrote rules with a letter, ran machines backwards and let the computer search. That is the start of Secondary 1 algebra, and you explained it in your own words. Well done, inquirers!"*

Reflection: What was hard? (1 minute)

SAY *"Last minute, just for you. On the last page of today's lesson, finish the three sentences in the box called "What was hard?": the hardest part was, I got unstuck by, next time I will try. Nobody marks it, and nobody has to show it. Then I would like to hear ONE sentence from two or three of you. "Pass" is fine."*

DO Run it at about minute 88, while you collect exit tickets (slide at the end). Never grade or correct it, and never read it out without the child's say-so. Praise the strategy, not the speed: say "you you tried a number and checked — that is what got you unstuck", never "you were quick". If a child writes "nothing was hard", ask "what would make it harder?".

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Extension for children who are ready

Build the search program in Scratch for 8n − 5 = 91. Then change it to look for n × n + n = 90, which has no easy backwards step. What does it say? (9)

Answer key (workbook)

Q1 (a) OUT for 10 is 33; rule OUT = 3n + 3 (b) IN is 10; rule OUT = 5n − 2 (c) OUT for 20 is 77; rule OUT = 4n − 3

Q2 (a) P gives 32, Q gives 23 (b) P: OUT = (n + 3) × 4; Q: OUT = 4n + 3 (c) No: (n + 3) × 4 = 4n + 12, so P is always 9 more than Q.

Q3 (a) n = 8 (43 + 5 = 48, 48 ÷ 6 = 8) (b) n = 10 (74 − 4 = 70, 70 ÷ 7) (c) n = 12 (45 ÷ 5 = 9, 9 + 3) (d) n = 24 (15 − 9 = 6, 6 × 4)

Q4 (a) 13 (b) −1 (c) 16 (25 + 7 = 32, 32 ÷ 2 = 16)

Q5 (a) Table: n = 1, 2, 3, 4; it says 5, 9, 13, 17 (b) 4n + 1 (c) 41

Q6 (a) set n to 0; repeat until (n * 8) − 5 = 91; change n by 1 (b) 12 (c) 91 + 5 = 96, 96 ÷ 8 = 12 ✓

Q7 (a) 7 gives 3; 10 gives 3 (b) n + 5, 2n + 10, 2n + 6, n + 3, 3 (c) Halving 2n + 6 gives n + 3; taking away n leaves 3, whatever n is.

Investigation IN 4, OUT 17: × 2 then + 9, × 3 then + 5, × 4 then + 1. (× 1 would need + 13, too big; × 5 would need −3, not allowed.) That is exactly three machines. Pattern: when the × number goes up by 1, the + number goes down by 4, because IN is 4 and the total must stay 17. IN 4, OUT 21: × 3 then + 9, × 4 then + 5, × 5 then + 1: again three machines, again down by 4. Look for: a table of × number and + number, a systematic order (try × 1, × 2, × 3, ...), and the explanation of the "down by 4". Stretch: a different IN gives a different step (IN 5 steps by 5).

Lesson 5: Every Possible Way

Week 3 (first lesson) • Counting • Workbook: Lesson 5

Objective	Count every possible case without missing one or counting one twice: list in order, use the multiplication rule for choices made one after another, take care with order and with 0 in digit puzzles, count handshakes and grid routes, and test each count with nested loops in code.
Thinking skills	Systematic listing, the multiplication rule, avoiding double counting, adding as you go (grid routes), tracing and planning nested loops
Materials	Laptop or desktop with Scratch 3 (scratch.mit.edu) open, the starter project "Every Order" (one loop that says each tea), mini whiteboards, digit cards 0 to 9, squared paper.
Online set-up	Children share their screen when they test code. The tutor shares the starter project as an .sb3 file in the chat (children use File, Load from your computer; no account needed), and the three starter blocks are also on the slide. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. Digit cards can be cut from paper. Breakout rooms in pairs for the Explore.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Welcome back, inquirers! Today's question sounds easy: how many ways? But counting every possible way, with none missing and none counted twice, is a skill that mathematicians and programmers use every day."*

"By the end of today you will make the computer list every case for you, and you will know when its answer is right."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

2–10 min Warm-up

DO Warm-up check first (slide 2, 3 minutes). It comes out of the 10-minute warm-up: play the game for about 3 minutes and ask only the first three quick questions after it.

SAY *"Two quick puzzles. They are new, not from last time. Write your answer AND one reason on your whiteboard. Hold it up when I say."*

Warm-up check	Answer and the reason to listen for
A café meal is 1 bread (3 kinds), 1 filling (4 kinds) and 1 drink (2 kinds). How many different meals are there?	24 meals: 3 × 4 × 2. Choices made one after another multiply. Tree or list: fix the bread, then the filling, then the drink.
5 chess clubs each play every other club once. How many games? Why is the answer not 5 × 4?	10 games. 5 × 4 = 20 counts every game twice (A against B is the same game as B against A), so halve it: 20 ÷ 2 = 10.

DO Fast track (nobody is labelled). If a child has BOTH answers right and a sensible reason, send a private chat message: “Today start on the Think Harder page at minute 40.” Fast-track children still join Explore, the mini-lesson and Talk time; only the worksheet time changes. Tell everyone: “Some of you start on a different page today. It changes every lesson, and you can move any time.” Never say “advanced”, never record the check as a mark, and let any child choose the main pages. A fast-track child who gets stuck goes back to the main page. This works because groups are small (up to 8) and each child has a printed workbook, so you can give children different pages quietly.

DO Count it, don’t list it. Six quick counting puzzles (pages 15 to 40, two-digit numbers, multiples of 5, lamp posts, numbers with a 7, even numbers). Teams race and must say how they counted.

SAY *“Six puzzles. Do not list them all: find a quick way to count, and be ready to say how you counted.”*

“First one: a book, pages 15 to 40. How many pages? Careful, it is not 25!”

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
3 × 4 × 2 = ?	24
1 + 2 + 3 + … + 10 = ?	55
How many whole numbers from 100 to 999?	900
6 × 5 ÷ 2 = ?	15
repeat 5 { repeat 2 { say hi } }: how many his?	10

10–25 min Explore (in pairs, with objects)

SAY *“The bubble tea shop has 3 teas and 2 toppings. Before you list them, write your guess: how many different orders? Now the shop adds sizes M and L. Will the number of orders go up by 2, or double, or something else? Write a prediction.”*

DO 1. A bubble tea shop has 3 teas (milk, green, taro) and 2 toppings (pearls, jelly).

2. On paper, list every order of 1 tea and 1 topping. How do you know you have them all?

3. The shop adds sizes M and L. Predict how many orders now, then check.

4. At the computer: open “Every Order” and put a second loop inside the first so the sprite says every order. (Teas are numbered 1 = milk, 2 = green, 3 = taro; toppings 1 = pearls, 2 = jelly. The sprite says join tea topping, so 21 means green with pearls.)

SAY *“Talk to your partner while you work. I will come round and ask you questions.”*

IF STUCK Stuck listing: write the first tea, then every topping with it. Then the next tea.

Code stuck: drag a second repeat block INSIDE the first one. Which loop should say the topping?

Finished early: add sugar levels 0%, 50% and 100%. Predict the number of orders first, then add a third loop to test it.

DO At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How many orders with 3 teas and 2 toppings? With sizes M and L too?	6 orders; with sizes, 12 (each of the 6 orders comes in 2 sizes: 6 × 2).
How did you make sure you had every order, with none twice?	Fix the tea, go through every topping, then move to the next tea: a list in order, like a tree.
In your code, which loop goes round faster?	The inner loop: it runs all the way round (2 toppings) for every single turn of the outer loop (each tea).

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"3 shirts and 2 pairs of shorts. I fix the red shirt and try every pair of shorts: red with black, red with grey. That is 2. Every shirt gets the same 2, so 3 shirts give 3 groups of 2: 3 × 2 = 6. Add 2 caps: each of the 6 outfits splits into 2, so 12."*

"In code, the outer loop picks the shirt and the inner loop picks the shorts. The inner loop goes all the way round for EACH shirt, so the say block runs 3 × 2 = 6 times."

"5 friends shaking hands. Friend A shakes 4 hands. B has already shaken A's, so 3 new ones. C: 2 new. D: 1 new. E: none left. 4 + 3 + 2 + 1 = 10. Or: 5 people times 4 hands is 20, but each handshake has two people in it, so I counted every one twice: 20 ÷ 2 = 10."

Worked example on the slides	Answer
The multiplication rule: 3 shirts and 2 pairs of shorts. Fix a shirt, then try every pair of shorts: 2 outfits for each shirt. 3 shirts give 3 groups of 2: 3 × 2. Add 2 caps: each outfit splits into 2.	6 outfits; with caps, 6 × 2 = 12.
Handshakes: 5 friends, every pair shakes once. A shakes 4 new hands, B 3 new, C 2, D 1, E 0. Or: 5 × 4 = 20 counts every handshake twice. In code: check every pair (a, b) and keep only a < b.	4 + 3 + 2 + 1 = 10, or 20 ÷ 2 = 10 handshakes.

SAY *"The key idea today is: every case, exactly once. Work in order: fix the first choice and run through every second choice. Choices made one after another multiply. Say it with me."*

DO Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Adding the choices instead of multiplying them (4 T-shirts and 3 shorts gives 7 outfits).
- Counting each handshake twice (7 × 6 = 42) or letting a person shake their own hand.
- Digit cards: putting 0 at the front of a number, or using the same card twice.

35–40 min Brain break

DO Every possible pose: arms up or arms out (2), and standing, squatting or on one leg (3). That is 2 × 3 = 6 poses. The tutor calls a number from 1 to 6 and everyone does that pose.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

40–75 min Worksheet (pairs, then alone, then stretch)

DO Minute 40: give out the workbook. Read the title of Lesson 5 aloud. Say "Pairs first. Talk before you write."

Minutes 40 to 55: pairs work together on the first page (with the short teaching slides). Minutes 55 to 72: each child works alone on the second page. Minutes 72 to 75: check together, then stretch and challenge for early finishers.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "Which choice did you fix first? How do you know you have them all?"
- Ask: "Point to the inner loop. How many times does it run for one turn of the outer loop?"
- Ask: "Have you counted any pair twice? Has anyone shaken their own hand?"

IF STUCK Hint 1: "Fix the first choice, then run through every second choice."

Hint 2: "Draw a tree, or a table with a row for each first choice."

Hint 3: "If the order does not matter, has your method counted each pair twice?"

DO Early finishers: give the stretch question first, then the challenge, then the Open investigation (it is the last box on the second Think Harder page). Fast-track children start there at minute 40.

Stretch and challenge questions

Question	Answer
Workbook Think Harder 1: A set lunch is 1 main, 1 drink and 1 dessert. There are 4 mains and 3 drinks. There are 60 different set lunches. How many desserts are there?	5 (4 × 3 = 12 and 60 ÷ 12 = 5)
Workbook Think Harder 2: How many 3-digit numbers can you make from the cards 1, 2, 3 and 4, using each card at most once? How many of them are even?	24 (4 × 3 × 2); 12 are even (the last card is 2 or 4: 2 choices, then 3 × 2 for the others)

Workbook Think Harder 3: In a league of 6 teams, every team plays every other team twice (once at home, once away). How many matches are played?	30 (6 × 5: each ordered pair is one home match)
Workbook Think Harder 4: A 4-digit PIN uses only the digits 1, 2 and 3. Digits may repeat. How many PINs are there?	81 (3 × 3 × 3 × 3)
Workbook Challenge 5: Make up your own “how many ways” puzzle whose answer is exactly 24. Write the puzzle, show how you counted, and check with a list or a tree. Swap with a partner.	Open. Examples: 4 children in a line (4 × 3 × 2 × 1 = 24); 2 shirts × 3 shorts × 4 caps = 24 outfits; 3-digit numbers from the cards 1, 2, 3, 4 using each card at most once (4 × 3 × 2 = 24). Check that every case is counted once.
Workbook Challenge 6: A set of dominoes has one domino for every pair of spot numbers from 0 to 6, including doubles such as 3 | 3. How many dominoes are in the set?	28 (7 doubles, plus 6 + 5 + 4 + 3 + 2 + 1 = 21 with two different numbers)
Workbook Challenge 7: How many 3-digit numbers have digits that add up to 5? (For example, 203.)	15 (first digit 1: 5 numbers, 2: 4, 3: 3, 4: 2, 5: 1; they are 104, 113, 122, 131, 140, 203, 212, 221, 230, 302, 311, 320, 401, 410, 500)
Workbook Open investigation (Find all the ways): Pay exactly 50 cents using only 5-cent, 10-cent and 20-cent coins (as many of each as you like). Find ALL the ways. Show how you made sure you did not miss one or count one twice. What if there is also a 50-cent coin?	There are 12 ways. Sort by the number of 20-cent coins. Two 20c: the other 10c is 10c or 5c + 5c: 2 ways. One 20c: the other 30c with 10c and 5c coins: 3×10, 2×10 + 2×5, 1×10 + 4×5, 6×5: 4 ways. No 20c: 50c with 10c and 5c coins: 5×10, 4×10 + 2×5, 3×10 + 4×5, 2×10 + 6×5, 1×10 + 8×5, 10×5: 6 ways. Total 2 + 4 + 6 = 12 (checked by brute force). With a 50c coin there is one more way, 13. Look for: an organised list (fix the biggest coin first), a table, no repeats, and a check. Do not accept a list that is only “some ways”.

These are the questions on the two Think Harder pages in the workbook. More are in the separate Question Bank.

75–82 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask two or three children to share, not only the fastest.

SAY *“Why does the multiplication rule work for outfits, but 5 × 5 gives the wrong answer for handshakes between 5 friends?”*

“How would you change a program that lists every pair so that it lists each handshake once?”

Discussion question	Listen for

Why does the multiplication rule work for outfits, but 5 × 5 gives the wrong answer for handshakes between 5 friends?	For outfits, every shirt goes with every pair of shorts and a shirt-and-shorts outfit is different from any other, so 3 × 2 counts each outfit once. For handshakes, 5 × 5 = 25 includes each person with themselves (5 cases) and counts A with B and B with A as two, so 25 − 5 = 20 and 20 ÷ 2 = 10. In code, keep only the pairs with a < b.

82–90 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Six friends meet. Every pair of friends shakes hands once. How many handshakes are there? Show how you know.	15 (5 + 4 + 3 + 2 + 1, or 6 × 5 ÷ 2).

SAY *"Today you counted things that would take ages to list: you fixed one choice, multiplied, and halved when you counted twice. Then the computer listed every case to test you. Notice it, test it, explain why."*

Reflection: What was hard? (1 minute)

SAY *"Last minute, just for you. On the last page of today's lesson, finish the three sentences in the box called "What was hard?": the hardest part was, I got unstuck by, next time I will try. Nobody marks it, and nobody has to show it. Then I would like to hear ONE sentence from two or three of you. "Pass" is fine."*

DO Run it at about minute 88, while you collect exit tickets (slide at the end). Never grade or correct it, and never read it out without the child's say-so. Praise the strategy, not the speed: say "you you listed them in order — that is what got you unstuck", never "you were quick". If a child writes "nothing was hard", ask "what would make it harder?".

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Extension for children who are ready

Write a program that counts every 3-digit number whose digits add up to 5 (three loops, one inside another). Then change 5 to other totals: which total gives the most numbers?

Answer key (workbook)

Q1 (a) 12 (4 × 3) (b) 24 (4 × 3 × 2) (c) 2 extra pairs: with 5 pairs, 4 × 5 × 2 = 40; with 4 pairs only 4 × 4 × 2 = 32.

Q2 (a) 9 (3 choices for the tens, as 0 cannot go first, then 3 for the ones: 3 × 3): 30, 35, 38, 50, 53, 58, 80, 83, 85 (b) 18 (3 × 3 × 2)

Q3 (a) a 1, b 1: 11; 1, 2: 12; 1, 3: 13; 1, 4: 14; 2, 1: 21; 2, 2: 22 (b) 12 (3 × 4) (c) 34 (d) 20 (5 × 4)

Q4 (a) 21 (6 + 5 + 4 + 3 + 2 + 1) (b) Every handshake has been counted twice, once for each person: 42 ÷ 2 = 21. (c) 10 people (9 + 8 + … + 1 = 45, or 10 × 9 ÷ 2 = 45)

Q5 (a) 20 (corner numbers, bottom row up: 1 1 1 1 / 1 2 3 4 / 1 3 6 10 / 1 4 10 20) (b) 8 (put 0 at P: 1 1 1 1 / 1 0 1 2 / 1 1 2 4 / 1 2 4 8). 12 routes go through P.

Q6 Gaps: repeat 7, repeat 7, if a < b (a > b also works: either keeps each pair once). (a) 21, the same as Q4(a); the "if" is checked 7 × 7 = 49 times. (b) It says 7: it counts each person with themselves (a = b), which is not a handshake.

Investigation There are 12 ways. Sort by the number of 20-cent coins. Two 20c: the other 10c is 10c or 5c + 5c: 2 ways. One 20c: the other 30c with 10c and 5c coins: 3×10, 2×10 + 2×5, 1×10 + 4×5, 6×5: 4 ways. No 20c: 50c with 10c and 5c coins: 5×10, 4×10 + 2×5, 3×10 + 4×5, 2×10 + 6×5, 1×10 + 8×5, 10×5: 6 ways. Total 2 + 4 + 6 = 12 (checked by brute force). With a 50c coin there is one more way, 13. Look for: an organised list (fix the biggest coin first), a table, no repeats, and a check. Do not accept a list that is only "some ways".

Lesson 6: Fair or Unfair?

Week 3 (second lesson) • Probability • Workbook: Lesson 6

Objective	Write a probability as a fraction (ways it can happen out of equally likely outcomes), list all 36 outcomes for two dice, decide whether a game is fair, change a game to make it fair, and run a game 1000 times in code, explaining why the result is close to what we expect but not exact.
Thinking skills	Listing equally likely outcomes, fractions in simplest form, expected number = fraction × number of tries, judging fairness, simulating with random numbers
Materials	Laptop or desktop with Scratch 3 (scratch.mit.edu), two dice per pair (or the tutor's dice program), a coin, coloured counters and a paper bag, mini whiteboards, tally sheet.
Online set-up	Children roll real dice at home; anyone without dice builds the dice program live in the shared Scratch window (pick random 1 to 6, twice) and uses that. This lesson is enrichment: probability is new to most children, so keep it playful. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. Pairs play the Explore game in breakout rooms and type their tallies into the chat so the group total can be added up.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Welcome, inquirers! Today we play games and then ask the big question: is this game fair? Our feelings are not enough. We will count outcomes with fractions, and then make the computer play a game a thousand times to test our maths."*

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

2–10 min Warm-up

DO Warm-up check first (slide 2, 3 minutes). It comes out of the 10-minute warm-up: play the game for about 3 minutes and ask only the first three quick questions after it.

SAY *"Two quick puzzles. They are new, not from last time. Write your answer AND one reason on your whiteboard. Hold it up when I say."*

Warm-up check	Answer and the reason to listen for
A spinner has 8 equal sections: 3 red and 5 blue. What is the chance of blue? Would "red wins / blue wins" be a fair game?	5/8. Not fair: red has 3/8, blue has 5/8, so blue wins more often. A fair game needs the same chance for both players.
Flip two coins. Ali wins if both coins match; Mei wins if they differ. Is that fair? List the outcomes.	Fair. The four outcomes are HH, HT, TH, TT. Match: HH and TT (2 out of 4). Differ: HT and TH (2 out of 4). Both 1/2.

DO Fast track (nobody is labelled). If a child has BOTH answers right and a sensible reason, send a private chat message: “Today start on the Think Harder page at minute 40.” Fast-track children still join Explore, the mini-lesson and Talk time; only the worksheet time changes. Tell everyone: “Some of you start on a different page today. It changes every lesson, and you can move any time.” Never say “advanced”, never record the check as a mark, and let any child choose the main pages. A fast-track child who gets stuck goes back to the main page. This works because groups are small (up to 8) and each child has a printed workbook, so you can give children different pages quietly.

DO Where on the line? Teams place six events on a 0 to 1 probability line and give each one’s fraction: roll a 7, flip heads, roll less than 5, pick a vowel from SINGAPORE, roll 1 to 6, pick the 1 red card from 10.

SAY *“Here is a line from 0 to 1. 0 means impossible, 1 means certain, 1/2 means an even chance.”*

“For each card, decide where it goes on the line and give its fraction. Teams, you have 20 seconds per card.”

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
Simplify 6/36	1/6
1/6 of 600	100
2/5 of 1000	400
Which is bigger: 5/12 or 4/9?	4/9 (16/36 is more than 15/36)
1 – 3/10	7/10

10–25 min Explore (in pairs, with objects)

SAY *“In the Dice Race, player A wins on six totals: 2, 3, 4, 10, 11, 12. Player B wins on only five: 5, 6, 7, 8, 9. Who do you think will win more often? Write A, B or “same”, and a reason.”*

DO 1. Roll two dice and add. A wins on 2, 3, 4, 10, 11 or 12. B wins on 5, 6, 7, 8 or 9.

2. Predict: who will win more often? A has 6 totals, B has only 5.

3. Play 20 rounds with your partner. Tally the wins.

4. Add up the whole group’s results. What do you notice?

SAY *“Talk to your partner while you work. I will come round and ask you questions.”*

IF STUCK Keep a tally: one column for A, one for B. Write the total each time.

Surprised by your results? Write down which totals came up most.

Finished early: in how many ways can two dice make a total of 7? Of 2? Use a red die and a blue die so 1 + 6 and 6 + 1 are different.

DO At about minute 22, say “Pencils down” and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
Who won more often in your pair? In the whole group?	Usually B. The group total is closer to B winning about 2 rounds in every 3.
A has more totals. Why does B win more often?	The totals are not equally likely: 7 can be made in 6 ways, 2 in only 1 way. B’s totals can be made in 24 of the 36 ways.
How could we be sure without playing forever?	List all 36 equally likely outcomes, or let a computer play 1000 games.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *“Roll one dice. More than 4 means 5 or 6: that is 2 ways out of 6 equally likely outcomes. So the probability is 2/6, which is 1/3.”*

“Two dice. The red die has 6 outcomes and for each one the blue die has 6, so 6 × 6 = 36, just like last lesson’s nested loop. A total of 7 happens 6 times: 1 + 6, 2 + 5, 3 + 4, 4 + 3, 5 + 2, 6 + 1. So P(7) = 6/36 = 1/6. A total of 2 happens once: 1/36.”

“In 1000 games I expect B to win about 2/3 × 1000, which is 666 2/3. A count cannot be 666 2/3, and every run is random, so the program will say something close, like 670 or 649, but almost never exactly 667.”

Worked example on the slides	Answer
A spinner with 6 equal parts: 3 blue, 2 yellow, 1 red. 6 equal parts, so 6 equally likely outcomes. Yellow: 2 parts out of 6. Spin 600 times: expect 1/3 of 600.	P(yellow) = 2/6 = 1/3; about 200 yellows in 600 spins (not exactly).
Two dice: the 36 outcomes. 6 outcomes for the red die × 6 for the blue = 36. Total 7: 1+6, 2+5, 3+4, 4+3, 5+2, 6+1. Explore game: A’s totals happen 12 ways, B’s 24 ways.	P(7) = 6/36 = 1/6. P(A wins) = 12/36 = 1/3, P(B wins) = 24/36 = 2/3: not fair.

SAY *“The key idea today is: count the equally likely outcomes. Probability = ways it can happen ÷ equally likely outcomes. A game is fair when every player has the same probability of winning. Say it with me.”*

DO Show the key idea slide at minutes 33 to 35. The “common mistakes” on that slide are for you, not the children. Watch for them during the worksheet:

- Treating outcomes as equally likely when they are not (two coins: “two heads, two tails, one of each” as thirds; dice totals 2 to 12 as elevenths).

- Thinking a random result should come out exactly as expected, or that a coin "is due" a head after several tails.
- Not simplifying, or writing the number of ways instead of the fraction (6 instead of 6/36 = 1/6).

35–40 min Brain break

DO Human dice. The tutor rolls a dice: 1 or 2, five star jumps; 3 or 4, touch your toes five times; 5 or 6, spin round once. Which will we do most? (Each has the same chance: 2/6 = 1/3.)

Water. Toilet break for anyone who needs it. Keep it to five minutes.

40–75 min Worksheet (pairs, then alone, then stretch)

DO Minute 40: give out the workbook. Read the title of Lesson 6 aloud. Say "Pairs first. Talk before you write."

Minutes 40 to 55: pairs work together on the first page (with the short teaching slides). Minutes 55 to 72: each child works alone on the second page. Minutes 72 to 75: check together, then stretch and challenge for early finishers.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "What are the equally likely outcomes here? How many are there?"
- Ask: "What fraction do you expect? So about how many out of 1000?"
- Ask: "Your program said 158, not 167. Is it broken? How could you check?" (run it again; run 10 000 games and look at the fraction)

IF STUCK Hint 1: "List every outcome. Are they all equally likely?"

Hint 2: "For two dice, draw the 6 by 6 table."

Hint 3: "Expected number = probability × number of tries."

DO Early finishers: give the stretch question first, then the challenge, then the Open investigation (it is the last box on the second Think Harder page). Fast-track children start there at minute 40.

Stretch and challenge questions

Question	Answer
Workbook Think Harder 1: A spinner has 10 equal parts numbered 1 to 10. What is the probability of spinning a prime number?	4/10 = 2/5 (2, 3, 5, 7)
Workbook Think Harder 2: Roll two dice and add. What is the probability that the total is 9?	4/36 = 1/9 (3+6, 4+5, 5+4, 6+3)
Workbook Think Harder 3: A bag has 5 red counters and some blue counters. The probability of picking red is 1/4. How many blue counters are there?	15 (5 out of 20 is 1/4)

Workbook Think Harder 4: Ali flips a coin 3 times. What is the probability that he gets heads all 3 times?	1/8 (8 equally likely outcomes: HHH, HHT, HTH, HTT, THH, THT, TTH, TTT)
Workbook Challenge 5: Design a FAIR game for two players using one dice, two coins or a spinner. Write the rules and show with a list or table that each player has the same chance of winning. Then change ONE rule to make the game unfair, and say who is favoured.	Open. Example: roll one dice; Player A wins on 1, 2 or 3 and Player B wins on 4, 5 or 6 (3/6 each). Change: A also wins on 4. Now A has 4/6 and B has 2/6, so A is favoured. Check that every outcome is listed and the winning outcomes add up to the whole.
Workbook Challenge 6: Mei and Ali each roll one dice. Mei wins if her number is bigger. Ali wins if his number is bigger or the same. Is the game fair? What is the probability that Mei wins?	Not fair. Mei wins 15/36 = 5/12; Ali wins 21/36 = 7/12 (15 bigger + 6 the same). Fair version: the same number means roll again.
Workbook Challenge 7: Spinner A has 3 equal parts: 1, 2, 3. Spinner B has 4 equal parts: 1, 2, 3, 4. Spin both and multiply. Even wins if the answer is even, Odd wins if it is odd. Is the game fair?	Not fair. There are 3 × 4 = 12 outcomes; the answer is odd only when both are odd: 2 × 2 = 4 ways. P(Odd) = 4/12 = 1/3, P(Even) = 8/12 = 2/3.
Workbook Open investigation (Is it always true? Test and explain): Ali says: “A game with only two outcomes is always fair.” Test him on three games. (1) Flip a coin: heads or tails. (2) Roll a dice: six or not six. (3) Roll two dice: total 7 or not 7. Which games are fair? Is Ali right? Change one unfair game to make it fair.	Ali is wrong. (1) Coin: 1/2 and 1/2, fair. (2) Dice: six has 1/6, not six has 5/6, unfair. (3) Two dice: a total of 7 has 6 out of 36 = 1/6, not 7 has 30 out of 36 = 5/6, unfair. Two outcomes is not the same as two EQUAL chances. Fair means every player has the same probability. Fixes: dice, 1 to 3 against 4 to 6; two dice, even total against odd total (18 out of 36 each, checked by listing all 36). Look for: a list or table of outcomes for game 3, probabilities as fractions, and the idea that one counter-example (game 2) is enough.

These are the questions on the two Think Harder pages in the workbook. More are in the separate Question Bank.

75–82 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask two or three children to share, not only the fastest.

SAY *“Mei flipped a coin 10 times and got 7 heads. She says the coin is unfair. Do you agree?”*

“How many flips would you need before you believed her? How could a program help?”

Discussion question	Listen for
Mei flipped a coin 10 times and got 7 heads. She says the coin is unfair. Do you agree? How could you test it?	Not yet: 10 flips is far too few. With a fair coin, 7 or more heads in 10 flips happens 176 times in 1024, about 1 time in 6. To test it, flip many more times (or simulate 1000 flips of a fair coin to see how much the counts wobble). If 1000 real flips gave 700 heads, that would be strong evidence: a fair coin almost never does that.

82–90 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Roll two dice and add. What is the probability that the total is 8? If you roll them 360 times, about how many 8s do you expect?	5/36 (2+6, 3+5, 4+4, 5+3, 6+2); about 50 (5/36 × 360 = 50).

SAY *"Today you measured chance with fractions, found the hidden unfairness in a game, and let the computer play a thousand games to test your maths. Close, not exact: that is how random things behave."*

Reflection: What was hard? (1 minute)

SAY *"Last minute, just for you. On the last page of today's lesson, finish the three sentences in the box called "What was hard?": the hardest part was, I got unstuck by, next time I will try. Nobody marks it, and nobody has to show it. Then I would like to hear ONE sentence from two or three of you. "Pass" is fine."*

DO Run it at about minute 88, while you collect exit tickets (slide at the end). Never grade or correct it, and never read it out without the child's say-so. Praise the strategy, not the speed: say "you you listed every outcome — that is what got you unstuck", never "you were quick". If a child writes "nothing was hard", ask "what would make it harder?".

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Extension for children who are ready

Change the Dice Race so that it is fair, using totals only (a total of 7 may mean "roll again"). Test your rules in code with 1000 games. How close to 500 : 500 do you get?

Answer key (workbook)

Q1 (a) 3/8 (b) 6/8 = 3/4 (c) about 375 (3/8 × 1000) (d) No. Each spin is random, so the count wobbles around 375; 362 is close. About 19 runs in 20 land between 345 and 405.

Q2 (a) about 500 (1/2 × 1000) (b) No. The flips are random, so each run gives a slightly different count near 500. Exactly 500 happens only about 1 run in 40. (c) about 200 (1/5 × 1000)

Q3 (a) 4/10 = 2/5 (b) 2 red (6 red, 6 blue: 6/12 = 1/2) (c) 10 blue (4 red out of 20: 4/20 = 1/5)

Q4 (a) Differences: 0 appears 6 times, 1: 10, 2: 8, 3: 6, 4: 4, 5: 2 (36 in all). (b) Ali 24/36 = 2/3, Ben 12/36 = 1/3 (c) Not fair: Ali is twice as likely to win, even though each has 3 differences. (d) Any split with 18 outcomes each, e.g. Ali on 1, 3, 5 (odd) and Ben on 0, 2, 4 (even); or Ali on 1, 2 and Ben on 0, 3, 4, 5; or Ali on 0, 1, 5 and Ben on 2, 3, 4.

Q5 Gaps: 6, 6, a = b. (a) About 167: P(double) = 6/36 = 1/6, and 1/6 × 1000 = 166 2/3. (b) A count must be a whole number, and each run is random, so it lands near 167 (about 19 runs in 20 between 145 and

190), not exactly on it.

Investigation Ali is wrong. (1) Coin: 1/2 and 1/2, fair. (2) Dice: six has 1/6, not six has 5/6, unfair. (3) Two dice: a total of 7 has 6 out of 36 = 1/6, not 7 has 30 out of 36 = 5/6, unfair. Two outcomes is not the same as two EQUAL chances. Fair means every player has the same probability. Fixes: dice, 1 to 3 against 4 to 6; two dice, even total against odd total (18 out of 36 each, checked by listing all 36). Look for: a list or table of outcomes for game 3, probabilities as fractions, and the idea that one counter-example (game 2) is enough.

Lesson 7: Turtle Geometry

Week 4 (first lesson) • Angles in Polygons • Workbook: Lesson 7

Objective	Use the idea that a turtle going once round a closed shape turns 360° altogether to find the turn angle (360° ÷ sides) and inside angle (180° – turn) of regular polygons, work backwards from an angle to the number of sides, extend to star polygons (turn = 360° × times round ÷ points), and test every prediction with a pen program in Scratch (or Python turtle).
Thinking skills	Turn (exterior) and inside (interior) angles, angles on a straight line, angle sum of a polygon, star polygons, predicting and tracing code, repeat loops and variables
Materials	A laptop or desktop with Scratch 3 open (scratch.mit.edu) and the Pen extension added, mini whiteboards, a protractor, plain or squared paper.
Online set-up	Pairs work in breakout rooms: one child drives Scratch and shares the screen, the other predicts, then they swap. Children hold drawings up to the camera. A tablet works for Scratch but is slower. Every drawing starts with the same 4 set-up blocks (erase all, go to x: -50 y: 100, point in direction 90, pen down) so shapes stay on the stage; turn blocks in Scratch show an arrow (turn ↻ is clockwise, our "turn right"). Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left (Motion, Pen, Control), not by colour.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Today you are turtle pilots. A turtle with a pen can draw any shape, but only if you tell it exactly how far to turn at every corner."*

"Our rule today: work the turn out with maths first, then test it in code. Notice it. Test it. Explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

2–10 min Warm-up

DO Warm-up check first (slide 2, 3 minutes). It comes out of the 10-minute warm-up: play the game for about 3 minutes and ask only the first three quick questions after it.

SAY *"Two quick puzzles. They are new, not from last time. Write your answer AND one reason on your whiteboard. Hold it up when I say."*

Warm-up check	Answer and the reason to listen for
A turtle goes once round a closed shape with equal turns of 40° and ends up facing the way it started. How many corners does the shape have?	9 corners. The turtle turns 360° altogether and each turn is 40°, so 360 ÷ 40 = 9.

A turtle goes round a closed shape with 4 corners. The first three turns are 90°, 120° and 60°. What is the fourth turn?	90°. Going once round, the turns add up to 360°. 90 + 120 + 60 = 270, and 360 − 270 = 90.

DO Fast track (nobody is labelled). If a child has BOTH answers right and a sensible reason, send a private chat message: “Today start on the Think Harder page at minute 40.” Fast-track children still join Explore, the mini-lesson and Talk time; only the worksheet time changes. Tell everyone: “Some of you start on a different page today. It changes every lesson, and you can move any time.” Never say “advanced”, never record the check as a mark, and let any child choose the main pages. A fast-track child who gets stuck goes back to the main page. This works because groups are small (up to 8) and each child has a printed workbook, so you can give children different pages quietly.

DO Turtle says. Stand and face the screen (north). The tutor calls turns; you turn your body and say which way you face. Last one: one full turn in 3 equal turns. How big is each?

SAY *“Stand up and face your screen. Call that north. I am the programmer and you are the turtle. Turn right 90. Which way are you facing now?”*

“Now: how can you make one full turn using 3 equal turns? How big is each turn?”

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
Angles on a straight line add up to …?	180°
How many degrees in a full turn?	360°
Two angles of a triangle are 50° and 75°. What is the third?	55°
What is 360 ÷ 8?	45
Face north. Turn right 90°, then right 180°. Which way do you face?	West

10–25 min Explore (in pairs, with objects)

SAY *“Here is the square program: repeat 4, move 100, turn right 90. I want an equilateral triangle. What turn should I type? Write your prediction on your whiteboard before anyone runs it.”*

DO 1. Add the Pen extension. Build the 4 set-up blocks, then repeat 4 [move 100, turn ↻ 90].

2. Change it to draw an equilateral triangle. Predict the turn first, then run it.

3. Now a regular pentagon (5 sides) and a regular hexagon (6 sides). Predict, then test.

4. Record sides, turn and “did it close?”. What do sides × turn always make?

SAY *“Talk to your partner while you work. I will come round and ask you questions.”*

IF STUCK Stuck: walk the shape yourself. Walk one side, then turn at the corner. How far did your body turn?

Typed turn 60 for the triangle? Run it and look. Does the turtle come home? How far has it turned altogether?

Finished early: predict the turn for a regular shape with 10 sides, then test it.

DO At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
What turn did the triangle need? What happened if you typed 60?	120°. Turn 60 draws only half a hexagon: 3 × 60 = 180°, so the turtle does not come home.
What turns did the pentagon and hexagon need?	Pentagon 72°, hexagon 60°.
What do sides × turn always make? Why?	360°: the turtle comes home facing the same way, so it has made one full turn.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"An octagon has 8 equal turns. To come home facing the same way the turtle turns 360° altogether, so each turn is 360 ÷ 8 = 45°. The turn and the inside angle sit on one straight line, so the inside angle is 180 − 45 = 135°."*

"Working backwards: the inside angle is 174°, so the turn is 180 − 174 = 6°. The turns make 360°, so there are 360 ÷ 6 = 60 sides."

"For the star the turtle goes round twice: 720° altogether. 5 equal turns: 720 ÷ 5 = 144°. The angle at each point is 180 − 144 = 36°."

Worked example on the slides	Answer
Regular octagon: find the turn. Going round once, the turtle turns 360° altogether. 8 equal turns: 360 ÷ 8 = ? Inside angle = 180° − turn. Test it: repeat 8 [move 80, turn ↻ ?]	Turn 45°, inside angle 135°.
A 5-point star. repeat 5 [move 150, turn ↻ 144] 5 × 144° = 720°: the turtle goes round twice. Turn = 360° × times round ÷ points = 720 ÷ 5. Angle at each point = 180° − turn = ?	Turn 144°, 36° at each point.

SAY *"The key idea today is: the 360° turn. Going once round a closed shape, the turtle turns 360° altogether. A regular polygon with n sides turns 360° ÷ n at each corner, and its inside angle is 180° − turn. Say it with me."*

DO Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Using the inside angle as the turn: turn 60° for a triangle draws half a hexagon, and turn 108° for a pentagon does not close.
- Dividing 180 by the number of sides instead of 360, or forgetting that turn + inside angle = 180°.
- Stars: thinking the turns always add up to 360°. A star goes round more than once, so its turns add up to 720°, 1080° and so on.

35–40 min Brain break

DO Turtle dance. Walk a square on the spot: step, quarter turn, four times. Then a triangle: step, turn a third of a turn, three times. Then a hexagon.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

40–75 min Worksheet (pairs, then alone, then stretch)

DO Minute 40: give out the workbook. Read the title of Lesson 7 aloud. Say "Pairs first. Talk before you write."

Minutes 40 to 55: pairs work together on the first page (with the short teaching slides). Minutes 55 to 72: each child works alone on the second page. Minutes 72 to 75: check together, then stretch and challenge for early finishers.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "What did you predict before you ran it? Were you right?"
- Ask: "How far has your turtle turned altogether? How do you know?"
- Ask: "Which angle does the turtle turn: the inside angle or the outside one? Show me on the picture."

IF STUCK Hint 1: "To come home facing the same way, the turtle makes one full turn: 360°."

Hint 2: "The corners are equal, so share 360° equally between them."

Hint 3: "The turn and the inside angle sit on one straight line: they add up to 180°."

DO Early finishers: give the stretch question first, then the challenge, then the Open investigation (it is the last box on the second Think Harder page). Fast-track children start there at minute 40.

Stretch and challenge questions

Question	Answer
Workbook Think Harder 1: Each inside angle of a regular polygon is 4 times its turn angle. How many sides does the polygon have?	10 (turn + 4 × turn = 180°, so the turn is 36° and 360 ÷ 36 = 10; check: inside 144° = 4 × 36°)
Workbook Think Harder 2: Each inside angle of a regular polygon is 140°. What do all its inside angles add up to?	1260° (turn 40°, so 9 sides; 9 × 140° = 1260°; check (9 − 2) × 180° = 1260°)

Workbook Think Harder 3: Mei's turtle draws a shape with 4 sides. Its first three turns are 70°, 100° and 90°. It comes home facing the same way after going round once. What is its last turn? What is the inside angle at that corner?	Last turn 100° (360 − 70 − 100 − 90); inside angle 80° (180 − 100)
Workbook Think Harder 4: A regular polygon has n sides. Write its turn angle and its inside angle using n.	Turn = 360 ÷ n; inside angle = 180 − 360 ÷ n
Workbook Challenge 5: Design your own star. Choose the number of points p and how many times it goes round, k, so that the turn angle 360° × k ÷ p is a WHOLE number and p and k share no common factor. Give the turn angle, the point angle (180° − turn) and the code. Explain why your star closes after p repeats.	Open. Example: 8 points going round 3 times: turn = 360 × 3 ÷ 8 = 135°, point angle 45°; code repeat 8 [move 100, turn ↻ 135]. After 8 repeats it has turned 8 × 135° = 1080° = 3 full turns, so it faces the way it started and the star closes. If p and k share a factor, the star closes sooner and is a smaller shape repeated (9 points going round 3 times is just a triangle).
Workbook Challenge 6: A square, a hexagon and a 12-sided shape fit exactly around a point: 90° + 120° + 150° = 360°. Find one other set of three regular polygons whose inside angles add up to 360° (hint: the answer uses a square or a triangle).	Square + octagon + octagon (90° + 135° + 135°) or triangle + 12-gon + 12-gon (60° + 150° + 150°).
Workbook Challenge 7: Try turn 120 and turn 60 with repeat 6 and move 100. Which one draws a hexagon? Why does the other not make a 6-point star?	Turn 60 draws the hexagon (6 × 60° = 360°, one full turn). Turn 120 draws a triangle twice round (6 × 120° = 720°, two full turns), and it only visits 3 corners, not 6.
Workbook Open investigation (What if we change ...? What stays the same?): A turtle repeats [move 100, turn ↻ t] until it first faces the way it started. Make a table for t = 40, 72, 90, 100, 120, 144 and 150: how many repeats, and how many times does it go round? Add up the turns to check. What changes when t changes? What stays the same?	Repeats and times round: 40: 9 and 1; 72: 5 and 1; 90: 4 and 1; 100: 18 and 5; 120: 3 and 1; 144: 5 and 2; 150: 12 and 5 (every row checked by brute force). Changes: the number of repeats and the shape (polygon or star). Stays the same: the turns always add up to a whole number of full turns, repeats × t = 360 × times round. When t divides 360 the turtle goes round once and draws a polygon; otherwise it goes round more than once and draws a star. Look for: adding the turns until the total is a multiple of 360, a table, and the link repeats = 360 × rounds ÷ t. Children do not need to test every row in Scratch: a few are enough.

These are the questions on the two Think Harder pages in the workbook. More are in the separate Question Bank.

75–82 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask two or three children to share, not only the fastest.

SAY *"Ali says the turtle turns by the inside angle of the shape. Mei says it turns by the outside angle. Who is right? Use a triangle to explain."*

"Why do the turns add up to 360° for every shape the turtle draws going round once?"

Discussion question	Listen for
Ali says the turtle turns by the inside angle of the shape. Mei says it turns by the outside angle. Who is right? Use a triangle to explain.	Mei is right. For an equilateral triangle the inside angle is 60° but the turtle turns 120°; turning 60° three times only draws half a hexagon. The turtle starts facing one way and comes home facing the same way after going round once, so it has made exactly one full turn: 360°. A star goes round more than once, so its turns add up to 720°, 1080° and so on.

82–90 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
A turtle draws a regular polygon and turns 5° at every corner. How many sides does the polygon have? What is each inside angle?	72 sides (360 ÷ 5 = 72); each inside angle is 175° (180 − 5)

SAY *"Today you predicted every turn with maths and then tested it in code. That is how mathematicians and programmers work together. Well done, inquirers!"*

Reflection: What was hard? (1 minute)

SAY *"Last minute, just for you. On the last page of today's lesson, finish the three sentences in the box called "What was hard?": the hardest part was, I got unstuck by, next time I will try. Nobody marks it, and nobody has to show it. Then I would like to hear ONE sentence from two or three of you. "Pass" is fine."*

DO Run it at about minute 88, while you collect exit tickets (slide at the end). Never grade or correct it, and never read it out without the child's say-so. Praise the strategy, not the speed: say "you you added up the turns — that is what got you unstuck", never "you were quick". If a child writes "nothing was hard", ask "what would make it harder?".

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Extension for children who are ready

Draw a rosette: repeat 12 [draw a square, then turn ↻ 30]. Start this one with go to x: 0 y: 0 instead of x: -50 y: 100, so the rosette stays on the stage. How far does the turtle turn altogether? (12 × (360 + 30) = 4680° = 13 full turns.) Design your own rosette and predict its total turning before you run it.

Answer key (workbook)

Q1 3 sides (example): 120°, 60°, 360°. 9 sides: turn 40°, inside 140°. 10 sides: 36°, 144°. 36 sides: 10°, 170°. Turn 24°: 15 sides, inside 156°. Inside 165°: turn 15°, 24 sides. Last column 360° every time. (a) Sides × turn = 360°: the turtle makes one full turn to come home facing the same way. (b) No: the turn would be 50°, and 360 ÷ 50 = 7.2 is not a whole number of sides.

Q2 (a) An equilateral triangle, drawn 3 times over (each turn of 120° is a triangle's turn, so after 3 moves the turtle is back where it started). (b) 9 × 120° = 1080° = 3 full turns. (c) Change the turn from 120 to 40 (360 ÷ 9).

Q3 Totals: 80, 160, 240, 320, 400, 480, 560, 640, 720. (a) 720 (b) 2 times (720 ÷ 360) (c) a 9-point star (d) 100° (180 – 80)

Q4 9 points, 4 times round: total 1440°, turn 160°, point 20°. 20 points, 3 times round: total 1080°, turn 54°, point 126°. Sharper points: the 9-point star (20°).

Q5 Each inside angle: 60°, 90°, 108°, 120°, 144°. Added: 180°, 360°, 540°, 720°, 1440°. (a) 180° (b) (n – 2) × 180°, or n × 180° – 360°

Investigation Repeats and times round: 40: 9 and 1; 72: 5 and 1; 90: 4 and 1; 100: 18 and 5; 120: 3 and 1; 144: 5 and 2; 150: 12 and 5 (every row checked by brute force). Changes: the number of repeats and the shape (polygon or star). Stays the same: the turns always add up to a whole number of full turns, repeats × t = 360 × times round. When t divides 360 the turtle goes round once and draws a polygon; otherwise it goes round more than once and draws a star. Look for: adding the turns until the total is a multiple of 360, a table, and the link repeats = 360 × rounds ÷ t. Children do not need to test every row in Scratch: a few are enough.

Lesson 8: Secret Codes

Week 4 (second lesson) • Clock (Modular) Arithmetic • Workbook: Lesson 8

Objective	Do clock arithmetic (counting past the end wraps round to the start), use n mod m as the remainder when n is divided by m, solve clock and calendar puzzles with mod 12 and mod 7, code and decode Caesar ciphers with letter numbers (A = 0 … Z = 25, code = (letter + shift) mod 26), crack a code, and test it all with a Scratch program that uses a list of letters and mod.
Thinking skills	Remainders and mod, clock and calendar arithmetic, negative numbers on a clock, Caesar ciphers, cracking codes by reasoning, lists and letters in code, tracing code in a table
Materials	A laptop or desktop with Scratch 3 open (the tutor shares a starter project with a list called alphabet holding A to Z), mini whiteboards, paper for code strips, scissors.
Online set-up	Children make their code strips on paper and hold them to the camera. Children hold up their coded name on paper to the camera, or the tutor reads out the coded names; partners decode on paper. Pairs use breakout rooms for the coding part; the driver shares the screen. Starter project: the tutor shares an .sb3 file (not a link only) containing the list alphabet with A to Z already filled in, and the loop already built; children add the three cipher blocks. Block colours on our slides are simplified; in Scratch, list blocks are red-orange (Variables palette) and you find blocks by the category name on the left, not by colour.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Today you are code-breakers. Spies, generals and computers all hide messages with codes. The oldest one we know is over 2000 years old, and it runs on the same maths as a clock."*

"Notice it. Test it. Explain why. Today "test it" means: crack it, then let the computer check."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

2–10 min Warm-up

DO Warm-up check first (slide 2, 3 minutes). It comes out of the 10-minute warm-up: play the game for about 3 minutes and ask only the first three quick questions after it.

SAY *"Two quick puzzles. They are new, not from last time. Write your answer AND one reason on your whiteboard. Hold it up when I say."*

Warm-up check	Answer and the reason to listen for
It is 8 o'clock. What time will it be 30 hours later, on a 12-hour clock? How do you know?	2 o'clock. 8 + 30 = 38, and 38 mod 12 = 2 (38 = 3 × 12 + 2). Or: 24 hours later is 8 again, then 6 more hours is 2.
Today is Wednesday. What day of the week will it be in 100 days?	Friday. 100 mod 7 = 2 (98 is 14 full weeks), so move 2 days on from Wednesday: Thursday, Friday.

DO Fast track (nobody is labelled). If a child has BOTH answers right and a sensible reason, send a private chat message: "Today start on the Think Harder page at minute 40." Fast-track children still join Explore, the mini-lesson and Talk time; only the worksheet time changes. Tell everyone: "Some of you start on a different page today. It changes every lesson, and you can move any time." Never say "advanced", never record the check as a mark, and let any child choose the main pages. A fast-track child who gets stuck goes back to the main page. This works because groups are small (up to 8) and each child has a printed workbook, so you can give children different pages quietly.

DO Clock jumps. The tutor shows a clock and calls a time and a jump ("9 o'clock, 4 hours later?"). Answer on whiteboards. Then jumps past 12, jumps backwards, and a day-of-the-week jump.

SAY *"Look at the clock. It is 9 o'clock. What time will it be 4 hours later? How did you work it out?"*

"Now 11 o'clock and 13 hours later. Did you go past 12? What happened?"

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
17 ÷ 5: what is the remainder?	2
30 ÷ 7: what is the remainder?	2
26 ÷ 26: what is the remainder?	0
40 ÷ 12: what is the remainder?	4
Which of 7, 9, 10, 12 leave remainder 1 when divided by 3?	7 and 10

10–25 min Explore (in pairs, with objects)

SAY *"With a shift of 3, A becomes D and B becomes E. What do you think Z becomes? Write your guess before you make the strip."*

DO 1. Make a code strip: write A to Z. Underneath, write the alphabet again starting at D (A → D, B → E ...).

2. Code your first name with shift 3. Swap with your partner and decode each other's.

3. Decode the tutor's message: PDWK LV IXQ

4. What happens to X, Y and Z? How do you decode: which way do you shift?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck making the strip: write A to Z on one line. Underneath, start at D under A, and keep going.

Stuck at the end of the alphabet: "After Z, where does the bottom row carry on?"

Finished early: invent a shift, code a short message for your partner, and give them only one clue.

DO At about minute 22, say “Pencils down” and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
What do X, Y and Z become with shift 3?	A, B and C: after Z the alphabet wraps round to the start.
How do you decode a shift-3 message?	Shift each letter back 3 (or forward 23). PDWK LV IXQ is MATH IS FUN.
What does a shift of 26 do?	Nothing: every letter goes once round and lands on itself.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *“It is 9 o’clock. I count on 5 hours: 10, 11, 12, 1, 2. Adding gives 14, but a clock wraps round after 12, so 14 − 12 = 2. In maths we write 14 mod 12 = 2: the remainder when 14 is divided by 12.”*

“To code Z with shift 5: Z is 25. 25 + 5 = 30. There is no letter 30, so I wrap round: 30 − 26 = 4, and 4 is E.”

“To decode A with shift 4: A is 0. 0 − 4 = −4. Below zero, so I add 26: −4 + 26 = 22, and 22 is W.”

Worked example on the slides	Answer
Clock arithmetic: 9 o’clock + 5 hours. Count on 5: 10, 11, 12, 1, 2. Adding gives 14. A clock wraps round after 12: 14 − 12 = 2. 14 mod 12 = 2: the remainder when 14 is divided by 12. Thursday + 45 days: 45 mod 7 = ?	2 o’clock. 45 mod 7 = 3, so Sunday.
Caesar code: ZOO with shift 5. Letter numbers: Z = 25, O = 14. Z: 25 + 5 = 30, 30 mod 26 = 4 = E. O: 14 + 5 = 19 = T. Decode AEB (shift 4): A = 0, 0 − 4 = −4, + 26 = 22 = W …	ZOO → ETT. AEB → WAX.

SAY *“The key idea today is: wrap round like a clock. n mod m is the remainder when n is divided by m. On a clock with m numbers, counting past the end wraps round to the start. Say it with me.”*

DO Show the key idea slide at minutes 33 to 35. The “common mistakes” on that slide are for you, not the children. Watch for them during the worksheet:

- Off by one: counting A as 1 in the maths (we use A = 0, Z = 25), or forgetting that Scratch’s list starts at item 1.
- Not wrapping round: writing a number bigger than 25 (or below 0 when decoding) instead of taking away or adding 26.

- Calendars: counting today as day 1 (Saturday + 2 days is Monday, not Sunday), or giving the quotient instead of the remainder for mod.

35–40 min Brain break

DO Clock arms. Your arm is the hour hand. The tutor calls a time and you point to it. Then "add 5 hours!" and you sweep your arm round to the new time. Try "add 12 hours!"

Water. Toilet break for anyone who needs it. Keep it to five minutes.

40–75 min Worksheet (pairs, then alone, then stretch)

DO Minute 40: give out the workbook. Read the title of Lesson 8 aloud. Say "Pairs first. Talk before you write."

Minutes 40 to 55: pairs work together on the first page (with the short teaching slides). Minutes 55 to 72: each child works alone on the second page. Minutes 72 to 75: check together, then stretch and challenge for early finishers.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "What number is that letter? Did you start at A = 0?"
- Ask: "Your number went past 25. What do you do?"
- Ask: "How can you check your code? Decode it again and see if you get back the message."

IF STUCK Hint 1: "Change each letter to its number first. Use the strip."

Hint 2: "Add the shift. If you go past 25, take away 26."

Hint 3: "To decode, take away the shift. If you go below 0, add 26."

DO Early finishers: give the stretch question first, then the challenge, then the Open investigation (it is the last box on the second Think Harder page). Fast-track children start there at minute 40.

Stretch and challenge questions

Question	Answer
Workbook Think Harder 1: The pattern CODECODECODE... goes on and on. What is the 100th letter?	E (100 mod 4 = 0, so the 100th letter is the 4th letter of CODE)
Workbook Think Harder 2: It is 10 o'clock now. What time will it be 1000 hours from now (on a 12-hour clock)?	2 o'clock (1000 mod 12 = 4, and 10 + 4 = 14, 14 − 12 = 2)
Workbook Think Harder 3: Find the smallest number bigger than 10 that leaves a remainder of 3 when it is divided by 4 and also when it is divided by 5.	23 (3 more than a multiple of 20: 23 = 5 × 4 + 3 = 4 × 5 + 3)

Workbook Think Harder 4: Decode BYFFI. It was made with a shift of 20.	HELLO (B = 1: 1 − 20 = −19, + 26 = 7 = H; Y = 24 − 20 = 4 = E; F = 5 − 20 = −15, + 26 = 11 = L; I = 8 − 20 = −12, + 26 = 14 = O)
Workbook Challenge 5: Write your own secret message of at least 8 letters and code it with a shift of your choice. Write clear instructions telling a friend how to decode it, and explain how a friend could crack it without being told the shift. Swap and decode.	Open. Check the code by decoding it, and that the cracking idea is sensible: try every shift from 1 to 25, or use the commonest letter (often E) or a short word (A or I). Example: MEET AT SIX with a shift of 3 is PHHW DW VLA.
Workbook Challenge 6: Ali codes a message with a shift of 7. Mei codes Ali's result with another shift. The final message is the same as the original coded with a shift of 2. What shift did Mei use?	21 (7 + 21 = 28, and 28 mod 26 = 2)
Workbook Challenge 7: Start at 12 on a clock and keep jumping forward 5 hours. Do you land on every number before you get back to 12? What about jumps of 4 hours?	Jumps of 5: yes, all 12 numbers (5, 10, 3, 8, 1, 6, 11, 4, 9, 2, 7, 12). Jumps of 4: no, only 4, 8 and 12, because 4 shares a factor with 12.
Workbook Open investigation (Is it always true? Test and explain): Code the word CAT with a shift of 8. Then code your answer with a shift of 18. What do you get? Try two other shifts that add up to 26. (a) Is it always true that two such shifts undo each other? Explain. (b) Is there a shift that is its own decoder, so that coding twice gives back the original? Find ALL of them.	CAT with shift 8 is KIB (C 2 → 10, A 0 → 8, T 19 → 27 mod 26 = 1). KIB with shift 18 is CAT again. (a) Yes: shifting by a and then by 26 − a moves every letter 26 places in total, a full circle round the alphabet, so each letter lands where it started. (b) Only shift 13: 13 + 13 = 26. No other shift from 1 to 25 works because 2 × shift would have to be a multiple of 26 (checked by brute force). Look for: the two worked examples, the "26 is a full circle" idea, and a systematic check for (b). Note 13 is the code known as ROT13.

These are the questions on the two Think Harder pages in the workbook. More are in the separate Question Bank.

75–82 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask two or three children to share, not only the fastest.

SAY *"Why does a shift of 26 change nothing?"*

"Ali says decoding shift 3 is the same as coding again with shift 23. Is he right? Why?"

Discussion question	Listen for
Why does a shift of 26 change nothing? Why is decoding a shift-3 message the same as coding it again with shift 23?	A shift of 26 moves every letter once round the whole 26-letter clock, so each letter lands back on itself (26 mod 26 = 0). Ali is right: shifting back 3 and shifting forward 23 land on the same letter, because 23 + 3 = 26 is a full turn. For example D (3): 3 − 3 = 0 = A, and 3 + 23 = 26, 26 mod 26 = 0 = A.

82–90 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Code the word HI with a shift of 25. What do you notice?	GH (H = 7, 7 + 25 = 32, 32 mod 26 = 6 = G; I = 8, 33 mod 26 = 7 = H). A shift of 25 is the same as shifting back 1, because 25 = 26 − 1.

SAY *"Today you used clock arithmetic to make and break codes, and you made the computer do it for you. Mathematicians call this modular arithmetic, and it helps protect messages sent on the internet. Well done, code-breakers!"*

Reflection: What was hard? (1 minute)

SAY *"Last minute, just for you. On the last page of today's lesson, finish the three sentences in the box called "What was hard?": the hardest part was, I got unstuck by, next time I will try. Nobody marks it, and nobody has to show it. Then I would like to hear ONE sentence from two or three of you. "Pass" is fine."*

DO Run it at about minute 88, while you collect exit tickets (slide at the end). Never grade or correct it, and never read it out without the child's say-so. Praise the strategy, not the speed: say "you you wrote out the alphabet — that is what got you unstuck", never "you were quick". If a child writes "nothing was hard", ask "what would make it harder?".

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Extension for children who are ready

The starter already keeps spaces (if the letter is in the alphabet list, code it, else join the letter itself). Add a "crack" button that says the message decoded with every shift from 1 to 25.

Answer key (workbook)

Q1 (a) 3 o'clock (15 − 12) (b) 5 o'clock (30 mod 12 = 6, 11 + 6 = 17, 17 − 12 = 5) (c) 7 o'clock (4 − 9 = −5, −5 + 12 = 7) (d) Monday (100 mod 7 = 2, Saturday + 2) (e) Thursday (30 mod 7 = 2, Saturday back 2) (f) 29 mod 12 = 5, 50 mod 7 = 1, 100 mod 26 = 22

Q2 (a) WPGGH (b) CODE IS FUN (c) 16 (10 + 16 = 26)

Q3 W: 22, 27, 1, B. A: 0, 5, 5, F. Y: 24, 29, 3, D. The program says BFD.

Q4 (a) BPM, probably THE (b) 8 (T = 19, 19 + 8 = 27, 27 mod 26 = 1 = B) (c) THE KEY IS UNDER THE MAT

Q5 (a) 15 (b) 5 (31 mod 26) (c) He gets the original message back: 9 + 17 = 26, a full turn (d) 13 (13 + 13 = 26)

Investigation CAT with shift 8 is KIB (C 2 → 10, A 0 → 8, T 19 → 27 mod 26 = 1). KIB with shift 18 is CAT again. (a) Yes: shifting by a and then by 26 – a moves every letter 26 places in total, a full circle round the alphabet, so each letter lands where it started. (b) Only shift 13: 13 + 13 = 26. No other shift from 1 to 25 works because 2 × shift would have to be a multiple of 26 (checked by brute force). Look for: the two worked examples, the “26 is a full circle” idea, and a systematic check for (b). Note 13 is the code known as ROT13.

Lesson 9: Data Detectives

Week 5 (first lesson) • Averages and Data • Workbook: Lesson 9

Objective	Find the mean as total ÷ number of values and as levelling out, use total = mean × number of values to find a missing value, predict how changing the data changes the mean, compare the mean with the median and mode when one value is extreme, and build a list program that adds up the data and finds the mean.
Thinking skills	Mean, working back from a mean to a total, reasoning about change, median and mode, lists and loops in code, tracing code
Materials	Laptop or desktop with Scratch 3 (scratch.mit.edu; free, in the browser), workbook, pencil, mini whiteboard, about 25 linking cubes or counters for levelling towers. A calculator only for checking.
Online set-up	Data is collected in the chat. Children share their screen to show their code. The tutor shares a starter project as an .sb3 file (one sprite with an empty list called data; children use File, Load from your computer) or children build it from the slide in 5 minutes. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour. The mean is familiar from primary school; the new parts are lists and loops, median and mode, and misleading averages.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

0–2 min Opening

SAY *"Welcome back, data detectives! Newspapers, games and even your PSLE results come with averages. Today we find out what the mean really is, how one number can trick it, and how a computer adds up a whole list in a blink."*

"Our rule today: work it out by hand first, then let the code test it. Notice it. Test it. Explain why."

DO Greet each child by name. Show the lesson title slide. Read "Today we will…" aloud.

2–10 min Warm-up

DO Warm-up check first (slide 2, 3 minutes). It comes out of the 10-minute warm-up: play the game for about 3 minutes and ask only the first three quick questions after it.

SAY *"Two quick puzzles. They are new, not from last time. Write your answer AND one reason on your whiteboard. Hold it up when I say."*

Warm-up check	Answer and the reason to listen for
The mean of 4 numbers is 10. Three of them are 6, 9 and 15. What is the fourth number?	10. The total must be 4 × 10 = 40. 6 + 9 + 15 = 30, and 40 − 30 = 10.

Five friends have pocket money $2, $3, $3, $4 and $38. The mean is $10. Is $10 a good description of what a friend typically has? Which average is better?	No: only one friend has more than $4. The median ($3) or the mode ($3) describes a typical friend better. One extreme value pulls the mean up.

DO Fast track (nobody is labelled). If a child has BOTH answers right and a sensible reason, send a private chat message: "Today start on the Think Harder page at minute 40." Fast-track children still join Explore, the mini-lesson and Talk time; only the worksheet time changes. Tell everyone: "Some of you start on a different page today. It changes every lesson, and you can move any time." Never say "advanced", never record the check as a mark, and let any child choose the main pages. A fast-track child who gets stuck goes back to the main page. This works because groups are small (up to 8) and each child has a printed workbook, so you can give children different pages quietly.

DO Quick-eye mean. The tutor flashes five numbers for 5 seconds. Write your best estimate of the mean, no working out. Closest guess wins a point. Three rounds.

SAY *"Quick-eye mean! I will show five numbers for five seconds only. No working out: just write your best guess of the mean. Closest guess wins a point."*

"How did you guess so fast? Who looked for a number in the middle and checked that the big ones and small ones balance?"

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
Find the mean of 4, 6 and 8.	6
Find the mean of 10, 20, 30 and 40.	25
5 children have a mean of 12 sweets. How many sweets altogether?	60
Find the mean of 3, 3, 3, 3 and 100.	22.4
Find the mean of 2.5 and 3.5.	3

10–25 min Explore (in pairs, with objects)

SAY *"Before we add anything up: the mean number of letters in our names. Will it be bigger or smaller than the number of letters in YOUR name? Write a prediction."*

"If our tutor joins with a 30-letter name, will the mean go up a little or a lot? Thumbs up for a lot, sideways for a little."

DO 1. Everyone posts in the chat the number of letters in their full name (no spaces). Write the list.

2. Estimate the mean. Then work it out by hand: total ÷ how many.

3. Type the data into the Scratch list called data and run the mean program. Same answer?

4. Predict, then test: add a value of 30 to the list. What happens to the mean?

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck on the hand calculation: add the values two at a time and write each running total, just like the variable total in the code.

Code says something strange: check that every value is in the list once, and that you divide by the length of the list, not by a number you typed.

Finished early: find a new value you could add that keeps the mean exactly the same. Test it in the program.

DO At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
Was your estimate close? How did you estimate?	Pick a value in the middle, then check that the values above it and below it roughly balance.
What happened to the mean when 30 was added?	It went up, a lot, because 30 is far above every other value. One big value pulls the mean towards it.
Did the program and your hand calculation agree?	They should. If not, look for a value typed twice or missed, or dividing by the wrong number.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Goals in 5 matches: 3, 0, 4, 1, 2. The 0 is a real value: that match counts. Total 3 + 0 + 4 + 1 + 2 = 10. Five matches, so the mean is 10 ÷ 5 = 2. Check by balancing: 3 is 1 above, 4 is 2 above: 3 above in all. 0 is 2 below, 1 is 1 below: 3 below in all. They balance, so 2 is right."*

"The mean of 4 test scores is 15, so the total must be 4 × 15 = 60. The three I know make 12 + 18 + 14 = 44. The missing score is 60 − 44 = 16. I never subtract from the mean: I subtract from the total."

"Goals by 7 players: 1, 2, 2, 3, 4, 5, 25. The mean is 42 ÷ 7 = 6, but 6 of the 7 players scored fewer than 6! The one big value pulled the mean up. The median, the middle value in order, is 3. That describes a typical player better."

Worked example on the slides	Answer
Find the mean, then balance-check. Goals in 5 matches: 3, 0, 4, 1, 2. Total: 3 + 0 + 4 + 1 + 2 = 10. There are 5 values (the 0 counts). Mean = 10 ÷ 5. Check: above the mean and below the mean should balance.	2. Above: +1 and +2. Below: −2 and −1. They balance.

Missing value: use the total. The mean of 4 test scores is 15. Three of them are 12, 18 and 14. Total of all 4: 4 × 15 = 60. Known: 12 + 18 + 14 = 44. Missing score = 60 − 44.	16. A 5th score of 15 would keep the mean at 15.

SAY *"The key idea today is: mean = total ÷ how many. Mean = total ÷ how many values. Turn it round: total = mean × how many. The values above and below the mean always balance. Say it with me."*

DO Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Dividing by the wrong count: leaving out a value of 0, or forgetting the new value when one is added.
- Finding a missing value from the MEAN instead of from the TOTAL (mean × how many) minus the known values.
- Thinking the mean must be one of the values, or that it is always a typical value even when one value is extreme.

35–40 min Brain break

DO Dice jumps. Roll two dice on screen: everyone does that many star jumps. Three rounds. What was the mean number of jumps per round?

Water. Toilet break for anyone who needs it. Keep it to five minutes.

40–75 min Worksheet (pairs, then alone, then stretch)

DO Minute 40: give out the workbook. Read the title of Lesson 9 aloud. Say "Pairs first. Talk before you write."

Minutes 40 to 55: pairs work together on the first page (with the short teaching slides). Minutes 55 to 72: each child works alone on the second page. Minutes 72 to 75: check together, then stretch and challenge for early finishers.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "What is the total? How many values are there? Did you count the 0?"
- Ask: "Before you calculate: will the mean go up, go down or stay the same? Why?"
- Ask: "Point to the block that adds up. What is total after the second time round the loop?"

IF STUCK Hint 1: "Find the total first. Mean × how many gives the total."

Hint 2: "Draw the values as towers. Move cubes from tall towers to short ones until they are level."

Hint 3: "For a trace table, do one loop at a time: which item? what is total now? then change i by 1."

DO Early finishers: give the stretch question first, then the challenge, then the Open investigation (it is the last box on the second Think Harder page). Fast-track children start there at minute 40.

Stretch and challenge questions

Question	Answer
Workbook Think Harder 1: The mean height of 4 children is 150 cm. A fifth child joins the group and the mean height becomes 152 cm. How tall is the fifth child?	160 cm (5 × 152 = 760, 4 × 150 = 600, 760 − 600 = 160)
Workbook Think Harder 2: Mei's mean score for 3 tests is 84. What must she score in the 4th test to raise her mean to 86?	92 (4 × 86 = 344, 3 × 84 = 252, 344 − 252 = 92)
Workbook Think Harder 3: Write 5 whole numbers with a mean of 6 and a mode of 4.	Open, e.g. 4, 4, 5, 8, 9 (total 30, and 4 appears most often). Check the total is 30.
Workbook Think Harder 4: A Scratch list holds 8 numbers with a mean of 12. Then the number 21 is added to the list. What will the mean program say now?	13 (8 × 12 = 96, 96 + 21 = 117, 117 ÷ 9 = 13)
Workbook Challenge 5: Make up your own "mean mystery" like Think Harder 8: choose five whole numbers, then write clues (any of mean, median, mode, smallest, largest) that give ONLY your numbers. Swap with a partner. Can they find a different set that fits your clues?	Open. Example clues: mean 5, median 4, mode 4, smallest 2, largest 9. The total is 25, so the other two numbers add to 25 − 2 − 4 − 9 = 10; only 4 and 6 fit (4 must appear twice to be the mode), giving 2, 4, 4, 6, 9. Check that the clues really give one set.
Workbook Challenge 6: The mean of 1, 2, 3, … , 99 is 50. Explain why without adding them all up. Then plan a short program that checks it.	Pair them: 1 + 99, 2 + 98, … , 49 + 51 each make 100, with 50 left in the middle. Every pair balances around 50, so the mean is 50 (total 4950 ÷ 99 = 50). Program: set total to 0, set n to 1, repeat 99 (change total by n, change n by 1), say total ÷ 99.
Workbook Challenge 7: A group of 8 children has a mean mass of 41 kg. When their tutor joins, the mean mass of all 9 becomes 44 kg. Then one more person joins and the mean of all 10 becomes 43 kg. What is the tutor's mass? What is the mass of the last person to join?	Tutor: 68 kg (9 × 44 = 396, 8 × 41 = 328, 396 − 328 = 68). Last person: 34 kg (10 × 43 = 430, 430 − 396 = 34).

Workbook Open investigation (What if we change ...? What stays the same?): Data: 4, 6, 7, 9, 14. Find the mean, the median and the mode (is there one?). Now change the 14 to 24, then change it to 4, then to 7. Make a table of the mean, median and mode each time. What changes? What stays the same? Can you change TWO numbers and keep the mean the same?	Original: mean 8, median 7, no mode. 14 to 24: mean 10, median 7, no mode. 14 to 4: mean 6, median 6, mode 4. 14 to 7: mean 6.6, median 7, mode 7 (all by computation). The mean changes every time because it uses every value. The median stayed 7 for 24 and 7 but fell to 6 for 4, because the order changed and a different number became the middle one. The median only cares about the middle; the mode appeared only when a value repeated. Two numbers: change one up by k and another down by k, for example 4 to 6 and 14 to 12; the total stays 40 and the mean stays 8. Look for: totals used, the middle found after re-ordering, and a reason for each change.

These are the questions on the two Think Harder pages in the workbook. More are in the separate Question Bank.

75–82 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask two or three children to share, not only the fastest.

SAY *"Can the mean be a number that nobody in the group has?"*

"Can the mean be bigger than every value in the list? Why or why not?"

Discussion question	Listen for
Can the mean be a number that nobody in the group has? Can the mean be bigger than every value in the list?	Yes: the mean of 2 and 3 is 2.5, a number nobody in the group has. No: the mean is always between the smallest and the largest value. Levelling out moves cubes from taller towers to shorter ones, so the level can never be higher than the tallest tower or lower than the shortest.

82–90 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
The mean of 3 numbers is 9. Two of them are 5 and 12. What is the third number? A fourth number, 13, is added. What is the new mean?	10 (total $3 \times 9 = 27$; $27 - 5 - 12 = 10$). New mean 10 ($27 + 13 = 40$, $40 \div 4 = 10$).

SAY *"Today you found means by levelling and by dividing, worked backwards from a mean to a total, and caught a misleading average. And your code added up a whole list for you. Next lesson is our Showcase Challenge: start thinking about a puzzle you would like to turn into a program."*

"Homework thought for next lesson: decide which starter you will change (A, B, C or Clock Jumper) and what your twist will be, so you can start building straight away."

Reflection: What was hard? (1 minute)

SAY *"Last minute, just for you. On the last page of today's lesson, finish the three sentences in the box called "What was hard?": the hardest part was, I got unstuck by, next time I will try. Nobody marks it, and nobody has to show it. Then I would like to hear ONE sentence from two or three of you. "Pass" is fine."*

DO Run it at about minute 88, while you collect exit tickets (slide at the end). Never grade or correct it, and never read it out without the child's say-so. Praise the strategy, not the speed: say "you you re-ordered the numbers — that is what got you unstuck", never "you were quick". If a child writes "nothing was hard", ask "what would make it harder?".

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Extension for children who are ready

Make a list of 5 whole numbers with mean 10, median 8 and mode 7 (one answer: 7, 7, 8, 13, 15). Then change your program so it also finds the largest value in the list.

Answer key (workbook)

Q1 (a) 30 (b) 6 (c) −2, +1, −4, +2, +3; they add up to 0 (the values above and below the mean balance)

Q2 (a) 68 points (17 × 4) (b) 19 (68 − 15 − 20 − 14) (c) 22 (5 × 18 = 90, 90 − 68 = 22)

Q3 (a) stays the same: 10 (60 ÷ 6) (b) goes up: 13 (c) goes down: 9 (36 ÷ 4) (d) goes up: 16 (80 ÷ 5)

Q4 (a) $22 (154 ÷ 7) (b) $15 (c) $12 (d) 6 pupils (e) Not fair: the one $69 pulls the mean up, and 6 of the 7 pupils get less than $22. The median $15 describes a typical pupil better.

Q5 Trace: i = 1, 2, 3, 4; item i = 5, 9, 6, 12; total = 5, 14, 20, 32. (a) 8 (32 ÷ 4) (b) repeat (length of scores) and say total ÷ (length of scores); it should now say 9 (45 ÷ 5)

Q6 (a) 31 °C (217 ÷ 7) (b) 3 days (33, 32 and 34) (c) Blank 1: mean. Blank 2: count. It says 3.

Investigation Original: mean 8, median 7, no mode. 14 to 24: mean 10, median 7, no mode. 14 to 4: mean 6, median 6, mode 4. 14 to 7: mean 6.6, median 7, mode 7 (all by computation). The mean changes every time because it uses every value. The median stayed 7 for 24 and 7 but fell to 6 for 4, because the order changed and a different number became the middle one. The median only cares about the middle; the mode appeared only when a value repeated. Two numbers: change one up by k and another down by k, for example 4 to 6 and 14 to 12; the total stays 40 and the mean stays 8. Look for: totals used, the middle found after re-ordering, and a reason for each change.

Lesson 10: Showcase Challenge

Week 5 (second lesson) • Mixed: Maths and Code • Workbook: Lesson 10

Objective	Use ideas from all nine lessons (sequences, primes, negative numbers, algebra, counting, probability, angles, clock arithmetic, averages) in a mixed challenge round, then plan, build, test and present a small puzzle program, working out the maths by hand first and explaining why it works.
Thinking skills	Choosing a strategy, connecting topics, planning code, testing with a known answer, presenting and explaining
Materials	Laptop or desktop with Scratch 3 (scratch.mit.edu), workbook (challenge round on page 1, planning sheet on page 2), pencil, mini whiteboard, the starter projects as .sb3 files (A Mystery Machine, B Prime or Not?, C Lucky Seven, D Clock Jumper; children use File, Load from your computer), or build them from the slides, certificates, stamps.
Online set-up	Challenge round in teams of 2 in breakout rooms. Building: everyone in the main room with cameras on, asking for help in the chat. Showcase: each child shares their screen for 3 minutes; the group gives two stars and a wish in the chat. Children may paste a screenshot into the chat instead and the tutor runs their file, which removes screen-share switching. Block colours on our slides are simplified; in Scratch, find blocks by the category name on the left, not by colour.
Prepare	Solve the worksheet yourself first. Set out one materials box per pair. Open the slides for this lesson and check the answers in the speaker notes.

DO Lesson 10 timing: this lesson changes the usual routine, so use this plan instead of the minute marks below. Warm-up 0 to 5; challenge round 5 to 22; starter projects 22 to 29; brain break 29 to 32; plan and build 32 to 62 (a working copy of one starter with one change is full marks); showcase 62 to 87 (3 minutes each plus a 20-second switch); exit ticket and certificates 87 to 90. The slide deck has the same times.

0–2 min Opening

SAY *"Welcome to our last lesson, inquirers! Two parts today. First a challenge round with puzzles from all nine lessons. Then you become the puzzle maker: you build a small program that sets a maths puzzle, and you show it to the group."*

"Our rule all program long: notice it, test it, explain why. Today you do all three in front of an audience."

DO Greet each child by name. Show the lesson title slide. Read "Today we will..." aloud.

2–10 min Warm-up

DO Warm-up check first (slide 2, 3 minutes). It comes out of the 10-minute warm-up: play the game for about 3 minutes and ask only the first three quick questions after it.

SAY *"Two quick puzzles. They are new, not from last time. Write your answer AND one reason on your whiteboard. Hold it up when I say."*

Warm-up check	Answer and the reason to listen for
The sequence 4, 9, 14, 19, … goes on. Is 2026 a term? How do you know?	No. The rule is 5n − 1, so 2026 + 1 = 2027 would have to be a multiple of 5, and it is not. (Every term ends in 4 or 9.)
Make 2-letter codes from A, B, C and D using different letters, where AB and BA count as different codes. How many codes are there?	12. 4 choices for the first letter and 3 for the second: 4 × 3 = 12. List: AB, AC, AD, BA, BC, BD, CA, CB, CD, DA, DB, DC.

DO Fast track (nobody is labelled). If a child has BOTH answers right and a sensible reason, send a private chat message: "Today start on the Think Harder page at minute 40." Fast-track children still join Explore, the mini-lesson and Talk time; only the worksheet time changes. Tell everyone: "Some of you start on a different page today. It changes every lesson, and you can move any time." Never say "advanced", never record the check as a mark, and let any child choose the main pages. A fast-track child who gets stuck goes back to the main page. This works because groups are small (up to 8) and each child has a printed workbook, so you can give children different pages quietly.

DO Lightning relay. The tutor asks a quick question from one of Lessons 1 to 9; whoever answers asks the next question (from the cards on screen) and names the next person.

SAY *"Lightning relay! I ask a quick question from one of our lessons, you answer, then you pass a new question to someone by saying their name. Ready? First one: what is the 10th term of 3, 7, 11, 15?"*

"Which lesson did that question come from? Can you remember the trick we learned?"

DO After the game, ask the quick questions on the warm-up slides. Use thumbs or a whisper to partners before taking answers.

Ask	Expect
What is the nth term of 4, 7, 10, 13, …?	3n + 1
Is 91 a prime number?	No: 7 × 13
−5 + 8 − 6 = ?	−3
A regular hexagon: what angle does the turtle turn at each corner?	60°
Mean of 7, 9, 11 and 13?	10

10–25 min Explore (in pairs, with objects)

SAY *"Look at the puzzles on page 1. Which one do you think will be hardest? Which lesson does each one come from? Write your guesses at the top of the page."*

DO 1. Teams of 2. Open the workbook at page 1: the challenge round.

2. Three puzzles from across the program. Maths first.

3. Score: 1 point for each answer, 1 for a check (a different way or in code), 1 for a clear reason.

4. After 12 minutes we check together and add up the points.

SAY *"Talk to your partner while you work. I will come round and ask you questions."*

IF STUCK Stuck on a puzzle: which lesson does it remind you of? Make a table, list it in order, or draw it.

Got an answer: earn the check point. Check it a different way, or test it in code.

Finished early: write a rule for Q1 that would make 2027 sticks. Is it possible with hexagons?

DO At about minute 22, say "Pencils down" and bring the group back together. Take the three questions below, one at a time.

Ask	Expect
How did you start the hexagon puzzle?	Make a table of hexagons and sticks; notice + 5 each time; the rule is 5n + 1.
How did you know you had found every way to write 40 as two primes?	Go through the primes in order up to 20 and test whether 40 minus each one is prime.
Which check did you do in code?	For example, run the hexagon loop, or list the digit-card numbers with nested loops and count the multiples of 3.

25–35 min Mini-lesson (you teach, they join in)

DO Show the mini-lesson slides. Think aloud as you work. Say what you notice, what you test and how you check. Children copy how you think.

SAY *"Hexagon chains: 6, 11, 16 sticks. Each new hexagon shares one side with the last one, so it adds 5 sticks. The first hexagon is 6 = 5 + 1. So n hexagons use 5n + 1 sticks. For 10 hexagons: 51. I can test that in code: start at 6 and change by 5 nine times."*

"Mystery Machine: the program picks n, works out 3n + 4 and shows it. If it shows 19, I go backwards: 19 − 4 = 15, 15 ÷ 3 = 5. Before I trust the program, I test it with a number I already know: if n is 5 it must show 19."

"Lucky Seven: there are 6 × 6 = 36 equally likely results for two dice. 1 + 6, 2 + 5, 3 + 4, 4 + 3, 5 + 2, 6 + 1 make 7: that is 6 of 36, one sixth. So in 1000 rolls I expect about 167. When I run it I got 152, then 182, then 162: close to 167 but never the same, because the rolls are random."

Worked example on the slides	Answer
Starter A: Mystery Machine. The program picks n from 1 to 20 and shows 3n + 4. The player works backwards: subtract 4, then divide by 3. Test with a value you know: n = 5 must show 19.	Out 25 → 7. Out 64 → 20. Out 3 is impossible: the smallest OUT for n from 1 to 20 is 7.

Starter C: Lucky Seven. Roll two dice 1000 times. Count the 7s. 36 equally likely results; 6 of them make 7. So expect about 1000 ÷ 6.	About 167. Each run gives a slightly different number (for example 152, 182, 162): random results vary.

SAY *"The key idea today is: maths first, then code to test it. Work out the answer by hand first. Then the program is a test: if it agrees with you, you are probably both right; if not, find out which one is wrong. Say it with me."*

DO Show the key idea slide at minutes 33 to 35. The "common mistakes" on that slide are for you, not the children. Watch for them during the worksheet:

- Coding before working out the maths, so there is no way to tell whether the output is right.
- Testing only once, or only with the value the program happened to show, instead of a value worked out by hand (and an edge case such as 1 or 0).
- Showcase: reading the code block by block instead of explaining the maths puzzle and why the program works.

35–40 min Brain break

DO Turtle moves. The tutor calls a polygon; everyone turns on the spot by its outside angle at every corner (triangle: 120°, square: 90°, hexagon: 60°) and counts until they face the front again.

Water. Toilet break for anyone who needs it. Keep it to five minutes.

40–75 min Worksheet (pairs, then alone, then stretch)

DO Minute 40: give out the workbook. Read the title of Lesson 10 aloud. Say "Pairs first. Talk before you write."

Minutes 40 to 55: pairs work together on the first page (with the short teaching slides). Minutes 55 to 72: each child works alone on the second page. Minutes 72 to 75: check together, then stretch and challenge for early finishers.

SAY *"Pairs first. Read each question aloud and talk about it before you write. I am not looking for speed. I am looking for good thinking."*

DO Circulate and ask:

- Ask: "What answer should your program give for this input? How do you know?"
- Ask: "What happens with an edge case: 1, 0, or the biggest number?"
- Ask: "In one sentence, what maths is your program about?"

IF STUCK Hint 1: "Start from a starter project and change one thing: the rule, the numbers, or the question."

Hint 2: "Write the maths on your planning sheet first. Then find one block for each line of your plan."

Hint 3: "If the program disagrees with you, trace it in a table: which value goes in, what does each block do?"

DO Early finishers: give the stretch question first, then the challenge, then the Open investigation (it is the last box on the second Think Harder page). Fast-track children start there at minute 40.

Stretch and challenge questions

Question	Answer
Workbook Think Harder 1: A row of squares uses 4, 7, 10, 13, … matchsticks. Which pattern number uses exactly 100 matchsticks?	Pattern 33 (3n + 1 = 100)
Workbook Think Harder 2: Find all the prime numbers between 80 and 100.	83, 89 and 97
Workbook Think Harder 3: A regular polygon has an angle of 150° inside each corner. How many sides does it have?	12 (the turn at each corner is 180° − 150° = 30°, and 360 ÷ 30 = 12)
Workbook Think Harder 4: Two dice are rolled and the numbers are added. What is the probability that the total is 10 or more?	6/36 = 1/6 (10: 3 ways, 11: 2 ways, 12: 1 way)
Workbook Challenge 5: Make up a challenge question that mixes TWO topics from the programme (for example primes and the clock, or sequences and negative numbers). Write the question, the answer and the working. Swap with a partner and solve each other's.	Open. Example: "The nth term of a sequence is 7n − 3. Which is the first term that is a multiple of 5?" The terms are 4, 11, 18, 25, so it is the 4th term, 25. Check that the answer is correct and that both topics are really needed.
Workbook Challenge 6: A spinner has the numbers 1 to 8, all equally likely. Ravi spins it twice and adds the numbers. What is the probability that the total is 9? Plan a program that tests it 1000 times.	8/64 = 1/8 (1 + 8, 2 + 7, … , 8 + 1), so about 125 in 1000. Program: set nines to 0; repeat 1000 (if pick random 1 to 8 + pick random 1 to 8 = 9 then change nines by 1); say nines.
Workbook Challenge 7: The nth term of a sequence is 6n − 1. Is 2027 (the year you start Secondary 1) in the sequence? Which is the first term bigger than 2027?	Yes: it is the 338th term (6 × 338 − 1 = 2027). The first term bigger than 2027 is the 339th, 2033.
Workbook Open investigation (Find all the ways): The sequence 4, 11, 18, 25, … has the rule 7n − 3. Find ALL the prime numbers in this sequence up to 100. Show your list of terms and how you tested each one. Then explain why no term of the sequence is a multiple of 7.	The terms up to 100 are 4, 11, 18, 25, 32, 39, 46, 53, 60, 67, 74, 81, 88, 95 (14 terms). The primes are 11, 53 and 67 (checked by brute force); the others are even or have a factor 3, 5 or 13 (25 = 5 × 5, 39 = 3 × 13, 81 = 9 × 9, 95 = 5 × 19). No term is a multiple of 7 because 7n is a multiple of 7, and 7n − 3 is 3 less, so dividing by 7 always leaves remainder 4. Look for: the full list, a test for every term using small primes, and the remainder reasoning. Stretch: change the rule to 7n + 1 and repeat.

These are the questions on the two Think Harder pages in the workbook. More are in the separate Question Bank.

75–82 min Talk time

DO Bring the group back to a circle (or camera on, mics on). Ask two or three children to share, not only the fastest.

SAY *"What was the hardest bug you fixed today? How did you find it?"*

"Which maths idea from the program would you like to explore further, and what would you try next?"

Discussion question	Listen for
Showcase: 3 minutes each. Show the puzzle, run the program, show one test where you knew the answer first, and explain the maths that makes it work.	Open answers. Listen for: testing with a value worked out by hand, tracing the code in a table, and maths links (nth term, primes, negative numbers, equations, probability, angles, averages). This is enrichment toward secondary maths and coding, not a preview of exam topics.

82–90 min Exit ticket and closing

DO Give each child a slip of paper (or ask them to write on their journal page). No talking, no helping. Show the exit ticket slide.

Exit ticket	Answer
Write one thing your program does, one test you did (the input and the answer you expected), and the maths that explains it.	Open. Look for a test worked out by hand before running the program, and a correct maths reason.

SAY *"Ten lessons ago you predicted the 100th term of a pattern and checked it in code. Today you built your own puzzle program and explained the maths to an audience. That is what inquirers do. Here is your certificate. Keep noticing, keep testing, and keep asking why!"*

"Give each child a certificate and tell them one specific thing they did well in their showcase."

Reflection: What was hard? (1 minute)

SAY *"Last minute, just for you. On the last page of today's lesson, finish the three sentences in the box called "What was hard?": the hardest part was, I got unstuck by, next time I will try. Nobody marks it, and nobody has to show it. Then I would like to hear ONE sentence from two or three of you. "Pass" is fine."*

DO Run it at about minute 88, while you collect exit tickets (slide at the end). Never grade or correct it, and never read it out without the child's say-so. Praise the strategy, not the speed: say "you you broke the problem into smaller parts — that is what got you unstuck", never "you were quick". If a child writes "nothing was hard", ask "what would make it harder?".

DO Stamp the inquiry journal when a child hands in their ticket. Mark S, D or H in the progress tracker at the back of this guide. If a child marked D or H, plan two minutes of catch-up at the start of the next lesson.

Extension for children who are ready

Level up your program: ask 5 questions and keep a score, let the player choose the rule, or count every dice total in a list. Write a Python version and check it gives the same results.

Answer key (workbook)

Q1 (a) 51 (b) 5n + 1 (c) 20 hexagons (5n + 1 = 101) (d) 76: it checks a chain of 15 hexagons (6 + 14 × 5)

Q2 (a) 49 (factors 1, 7, 49; the only other 2-digit number with exactly 3 factors is 25) (b) 2 × 2 × 3 × 7 (c) 3 + 37, 11 + 29, 17 + 23 (3 ways)

Q3 (a) −7 °C (−4 + 9 = 5, 5 − 12 = −7) (b) 5 (5 × 3 − 10 = 5) (c) 15 sides (360 ÷ 24) (d) 1 o'clock (9 + 100 = 109 = 9 × 12 + 1) (e) 12 numbers; 6 even (f) 4/12 = 1/3 (12, 21, 24, 42)

Q4 (a) 31; 18 ((58 − 4) ÷ 3) (b) 4 factors (1, 7, 13, 91): not prime (c) about 139: 5 of the 36 equally likely results make 8 (2 + 6, 3 + 5, 4 + 4, 5 + 3, 6 + 2), and 1000 × 5 ÷ 36 ≈ 139; the program gives a number near this, not exactly

Q5 Open: an idea, the puzzle, its answer worked out by hand, and a twist.

Q6 Open: blocks (under 15) and at least two tests worked out before running, one of them an edge case.

Investigation The terms up to 100 are 4, 11, 18, 25, 32, 39, 46, 53, 60, 67, 74, 81, 88, 95 (14 terms). The primes are 11, 53 and 67 (checked by brute force); the others are even or have a factor 3, 5 or 13 (25 = 5 × 5, 39 = 3 × 13, 81 = 9 × 9, 95 = 5 × 19). No term is a multiple of 7 because 7n is a multiple of 7, and 7n − 3 is 3 less, so dividing by 7 always leaves remainder 4. Look for: the full list, a test for every term using small primes, and the remainder reasoning. Stretch: change the rule to 7n + 1 and repeat.

Progress tracker

After each exit ticket, write S (secure: correct and can explain), D (developing: correct but cannot explain yet) or H (needs help). Use the notes for what to revisit next lesson.

Student	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	Notes

Ending the program

- In Lesson 10, give each child a certificate and their completed inquiry journal.
- Send parents a short message through the approved channel: one thing their child did well, and one thing to keep practising.
- Delete or hand over all Student Data within 7 days after the last lesson (this is in the tutor agreement).

Tutor checklist

Before each lesson

- Read the lesson plan and solve the worksheet yourself without looking at the key.
- Check materials are ready, or the online set-up has been sent.
- Test the platform, screen share and annotation tools 10 minutes early (online).
- Print or open the slide deck for the lesson.

During the lesson

- Circulate and ask questions rather than giving answers.
- Use only the approved channels to contact parents. No private messages to children.
- Do not take photos or recordings unless the Company has approved it and parents have consented.
- If you have a concern about a child's safety or welfare, tell the Company straight away.

After the lesson

- Stamp inquiry journals and fill in the tracker.
- Collect worksheets and answer keys (face to face) and keep them secure.
- Note any child who needs help, and plan a short catch-up at the start of the next lesson.

Math Inquirers Holiday Program

Notice it. Test it. Explain why.

A 10-lesson holiday maths program for Primary 6 students after the PSLE

Your child will spend the holidays as a young mathematician: noticing patterns, testing ideas with real objects and explaining their thinking. Lessons are live online in small groups, hands-on, and built around thinking, not speed.

What your child will do

- Lesson 1, Code the Pattern: sequences and the nth term
- Lesson 2, Prime Hunters: primes, factors and prime factorisation
- Lesson 3, Below Zero: negative numbers
- Lesson 4, Algebra Machines: expressions and equations
- Lesson 5, Every Possible Way: counting
- Lesson 6, Fair or Unfair?: probability
- Lesson 7, Turtle Geometry: angles in polygons
- Lesson 8, Secret Codes: clock (modular) arithmetic
- Lesson 9, Data Detectives: averages and data
- Lesson 10, Showcase Challenge: mixed: maths and code

How you can help

- Ask “How do you know?” when your child tells you an answer.
- Let your child try first. Please sit nearby, but let them find the answer.
- Print the workbook and cut-outs, and keep scissors, a ruler and a pencil ready.

Practical details

Dates	[start date] to [end date], two lessons a week
Time	[day] and [day], [start time] to [end time]
Where	Live online video class (link sent after registration)
Group size	[number] children
Fee	S$500 introductory (regular S$650), workbook included, payable in 3 instalments
Contact	[name, phone or email]

Math Inquirers Holiday Program is an independent enrichment program. It is inspired by gifted-education-style thinking skills but is not affiliated with, run by or endorsed by the Ministry of Education or any school.